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INTRODUCTION

The differences which distinguish immunocompetent T- and B-cells are sharply demarcated at the cell surface (table 9.1). The most clear-cut discrimination is established by reagents which recognize components of the antigen receptors, anti-CD3 for T-cells and anti-immunoglobulin for B-cells, and in laboratory practice these are the markers most often used to enumerate the two lymphocyte populations. The surface expression of receptors for Ig and complement also distinguishes T- and B-cells, while the adventitious formation of rosettes through the binding of sheep erythrocytes to CD2 is sometimes used for the detection and separation of T-cells.

Differences in the CD markers determined by monoclonal antibodies reflect disparate functional properties and in particular define specialized T-cell subsets. CD4 is a marker of T-helper cell populations which promote activation and matu-

ration of B-cells and cytotoxic T-cells, and control antigen-specific chronic inflammatory reactions through stimulation of macrophages. CD4 molecules form links with class II *major histocompatibility complex* (MHC) on the cell presenting antigen. Similarly, the CD8 molecules on cytotoxic T-cells associate with MHC class I (figure 9.1).

Attention should also be drawn to the 'nonspecific mitogens' which activate populations and sometimes subpopulations of T- or B-cells in a way which is unrelated to the antigen specificity of the lymphocyte receptors. They react with constant, as distinct from highly variable, structures on the cell surface. For this reason, they are often termed polyclonal B- or T-cell activators (table 9.1). We have already drawn attention in Chapter 5 to the ability of superantigens such as staphylococcal enterotoxins to act as polyclonal activators by stimulating all T-cells bearing certain T-cell receptor (TCR) V β families irrespective of their specificity for antigen.

T-LYMPHOCYTES AND ANTIGEN-PRESENTING CELLS INTERACT THROUGH SEVERAL PAIRS OF ACCESSORY MOLECULES

The affinity of an individual TCR for its specific MHC-antigen peptide complex is relatively low (figure 9.2) and a sufficiently stable association with the antigen-presenting cell (APC) can only be achieved by the interaction of complementary pairs of accessory

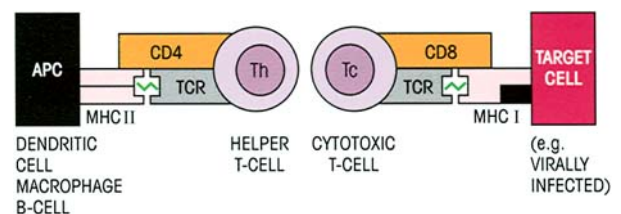


Figure 9.1. Helper and cytotoxic T-cell subsets are restricted by MHC class. CD4 on helper T-cells contacts MHC class II; CD8 on cytotoxic T-cells associates with class I.

Table 9.1. Comparison of human T- and B-cells.

	T-cells	B-cells
% in peripheral blood	65–80	8–15
ANTIGEN RECOGNITION	Processed	Native
CELL SURFACE MOLECULES		
Antigen receptor	TCR/CD3	Surface Ig
MHC class I	+	+
MHC class II	only after activation	+
CD2	+	–
CD4	MHC class II-restricted (helper)	–
CD5	+	only on B1a minor subset
CD8	MHC class I-restricted (cytotoxic)	–
CD19	–	+
CD20	–	+
CD21 (CR2: C3d and EBV receptor)	–	+
CD23 (FcεRII)	–	+
CD32 (FcγRII)	–	+
POLYCLONAL ACTIVATION	Anti-CD3 Phytohemagglutinin Pokeweed mitogen * <i>S. aureus</i> enterotoxin	Anti-Ig Epstein-Barr virus Pokeweed mitogen <i>S. aureus</i> Cowan I strain

*Staphylococcal superantigen, polyclonal response restricted to certain TCR Vβ families.

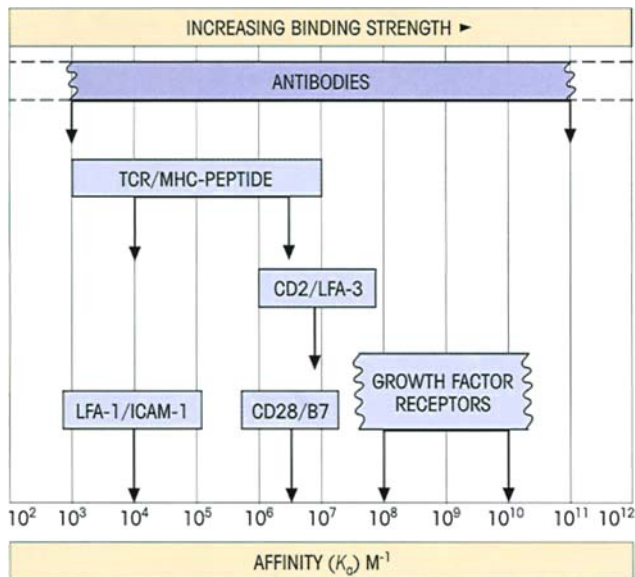


Figure 9.2. The relative affinities of molecular pairs involved in interactions between T-lymphocytes and cells presenting antigen. The ranges of affinities for growth factors and their receptors, and of antibodies, are shown for comparison. (Based on Davies M.M. & Chien Y.-H. (1993) *Current Opinion in Immunology* 5, 45.)

molecules such as LFA-1/ICAM-1, CD2/LFA-3, and so on (figure 9.3). However, these molecular couplings are not necessarily concerned just with intercellular adhesion.

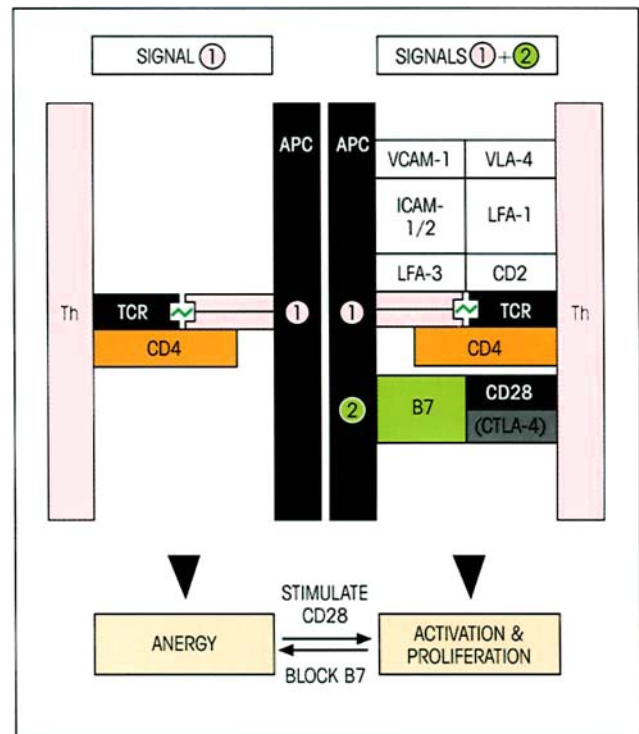


Figure 9.3. Activation of resting T-cells. Interaction of costimulatory molecules leads to activation of resting T-lymphocyte by antigen-presenting cell (APC) on engagement of the T-cell receptor (TCR) with its antigen–MHC complex. Engagement of the TCR signal 1 without accompanying costimulatory signal 2 leads to anergy. Note, a cytotoxic rather than a helper T-cell would, of course, involve coupling of CD8 to MHC I. Engagement of CTLA-4 with B7 downregulates signal 1. ICAM-1/2, intercellular adhesion molecule-1/2; LFA-1/2, lymphocyte function-associated molecule-1/2; VCAM-1, vascular cell adhesion molecule-1; VLA-4, very late antigen-4. (Based on Liu Y. & Linsley P.S. (1992) *Current Opinion in Immunology* 4, 265.)

THE ACTIVATION OF T-CELLS REQUIRES TWO SIGNALS

Antibodies to the TCR, either anti-idiotypic or anti-CD3, when insolubilized by coupling to Sepharose, will not fully activate resting helper T-cells on their own. Upon addition of interleukin-1 (IL-1), however, RNA and protein synthesis is induced, the cell enlarges to a blast-like appearance, interleukin-2 (IL-2) synthesis begins and the cell moves from G₀ into the G₁ phase of the mitotic cycle. Thus, two signals are required for the activation of a resting helper T-cell (figure 9.3). Antigen in association with MHC class II on the surface of APCs is clearly capable of fulfilling these requirements. Complex formation between the TCR, antigen and MHC provides signal 1, through the receptor–CD3 complex, and this is greatly enhanced

by the coupling of CD4 with the MHC. The T-cell is now exposed to a costimulatory signal 2 from the APC. Although this could be IL-1, it would appear that the most potent costimulator is B7 on the APC binding to CD28. Thus activation of resting T-cells can be blocked by anti-B7; surprisingly, this renders the T-cell **anergic**, i.e. unresponsive to any further stimulation by antigen. As we shall see in later chapters, the principle that two signals activate, but one may induce anergy in, an antigen-specific cell provides a potential for targeted immunosuppressive therapy. Unlike resting T-lymphocytes, **activated T-cells proliferate in response to a single signal**.

Adhesion molecules such as ICAM-1, VCAM-1 and LFA-3 are not themselves costimulatory but augment the effect of other signals (figure 9.3); an important distinction. Early signaling events involve the aggregation of **lipid rafts** composed of membrane subdomains enriched in cholesterol and glycosphingolipids. The cell membrane molecules involved in activation become concentrated within these structures.

PROTEIN TYROSINE PHOSPHORYLATION IS AN EARLY EVENT IN T-CELL SIGNALING

If a **protein tyrosine kinase (PTK)** phosphorylates and thereby activates a kinase precursor, which in turn switches on a second kinase precursor and so on, one has the basis for an enzymic phosphorylation cascade which could amplify an initial signal, just as the proteolytic enzyme cascade amplifies the triggering event of the complement system (cf. p. 10). As we shall see shortly, these signaling cascades can become quite extensive! (figure 9.4); but take it one step at a time and you'll be O.K.

The initial signal for T-cell activation through the TCR is greatly enhanced by cross-linking the TCR with CD4 which brings the CD4-associated PTK, **Lck**, close to the CD3-associated **ζ chains**. Immunoreceptor tyrosine-based activation motifs (ITAMs) on the ζ chains become phosphorylated and bind to the SH2 domains of **ZAP-70**, which now becomes an active PTK (figure 9.5) capable of initiating a series of downstream biochemical events. The key role of these tyrosine kinases is underlined by the ability of the PTK inhibitor herbimycin-A to block proximal TCR-mediated signaling events such as phosphatidylinositol turnover as well as later manifestations like IL-2 production. Receptor proximal events are coupled to

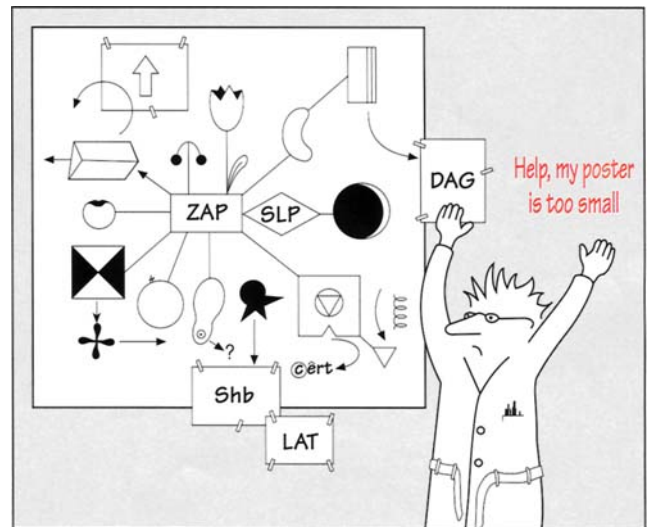


Figure 9.4. Signaling pathways can become quite complex. (Reproduced with permission from Zolnierowicz S. & Bollen M. (2000) *EMBO Journal* 19, 483.)

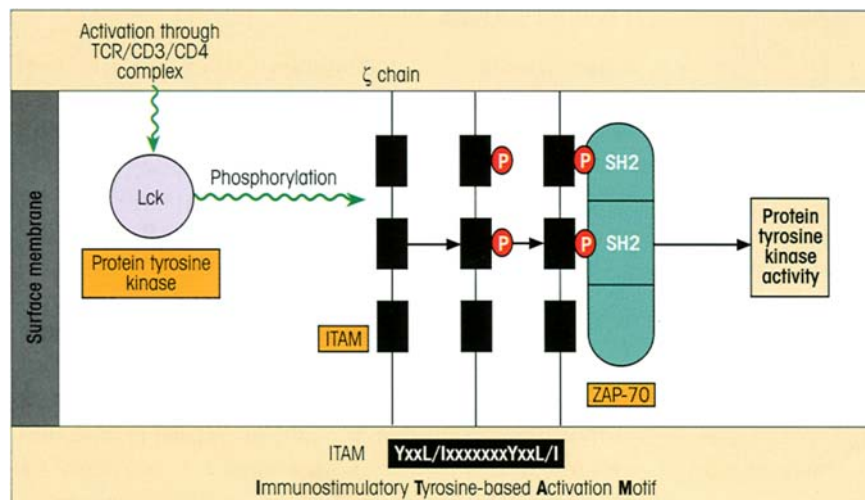


Figure 9.5. Signals through the TCR/CD3/CD4/8 complex initiate a protein tyrosine kinase (PTK) cascade. The PTK Lck (p56^{lck}) phosphorylates the tyrosine within the ITAM sequences of CD3-associated ζ chains. These bind the ζ-associated protein (ZAP-70) through its SH2 (Src homology-2) domains and this in turn acquires PTK activity for downstream phosphorylation of later components in the chain. In contrast to the three ITAMs on each ζ chain, the CD3 γ, δ and ε chains each bear a single ITAM.

downstream signaling cascades by nonenzymic **adaptor proteins** which link together the enzymes necessary for signal transduction.

DOWNSTREAM EVENTS FOLLOWING TCR SIGNALING

Ras function

Following TCR signaling, there is an early increase in the level of active Ras–GTP (guanosine triphosphate) complexes which regulate pivotal **mitogen-activated protein kinases (MAPK)**, such as JNK or ERK, through sequential kinase cascades. Thus, in one of the phosphorylation amplifying cascades, MEK (MAP-ERK kinase) acts as a MAP kinase kinase and Raf as a MAP kinase kinase kinase or MAPKKK (figure 9.6)! As illustrated in figure 9.7, this is just one of a number of

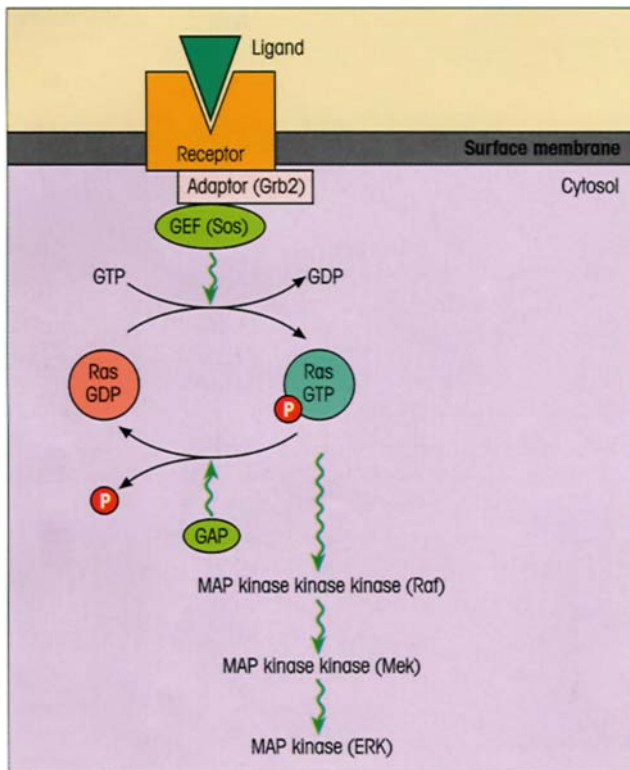


Figure 9.6. Regulation of Ras activity controls kinase amplification cascades. A number of cell surface receptors signal through Ras-regulated pathways. Ras cycles between inactive Ras–GDP and active Ras–GTP, regulated by guanine nucleotide exchange factors (GEFs) which promote the conversion of Ras–GDP to Ras–GTP, and by GTPase-activating proteins (GAPs) which increase the intrinsic GTPase activity of Ras. Upon ligand binding to receptor, receptor tyrosine kinases recruit adaptor proteins, e.g. Grb2, and GEF proteins, such as Sos ('son of sevenless'), to the plasma membrane. These events generate Ras–GTP which activates Raf. (Modified with permission from Olson M.F. & Marais R. (2000) *Seminars in Immunology* 12, 63.)

different pathways involved in T-cell activation. The MAP kinases themselves may also be influenced by CD28 operating through phosphatidylinositol 3-kinase (PI3K). It should be stressed that the details of these various pathways are not yet inscribed in tablets of stone.

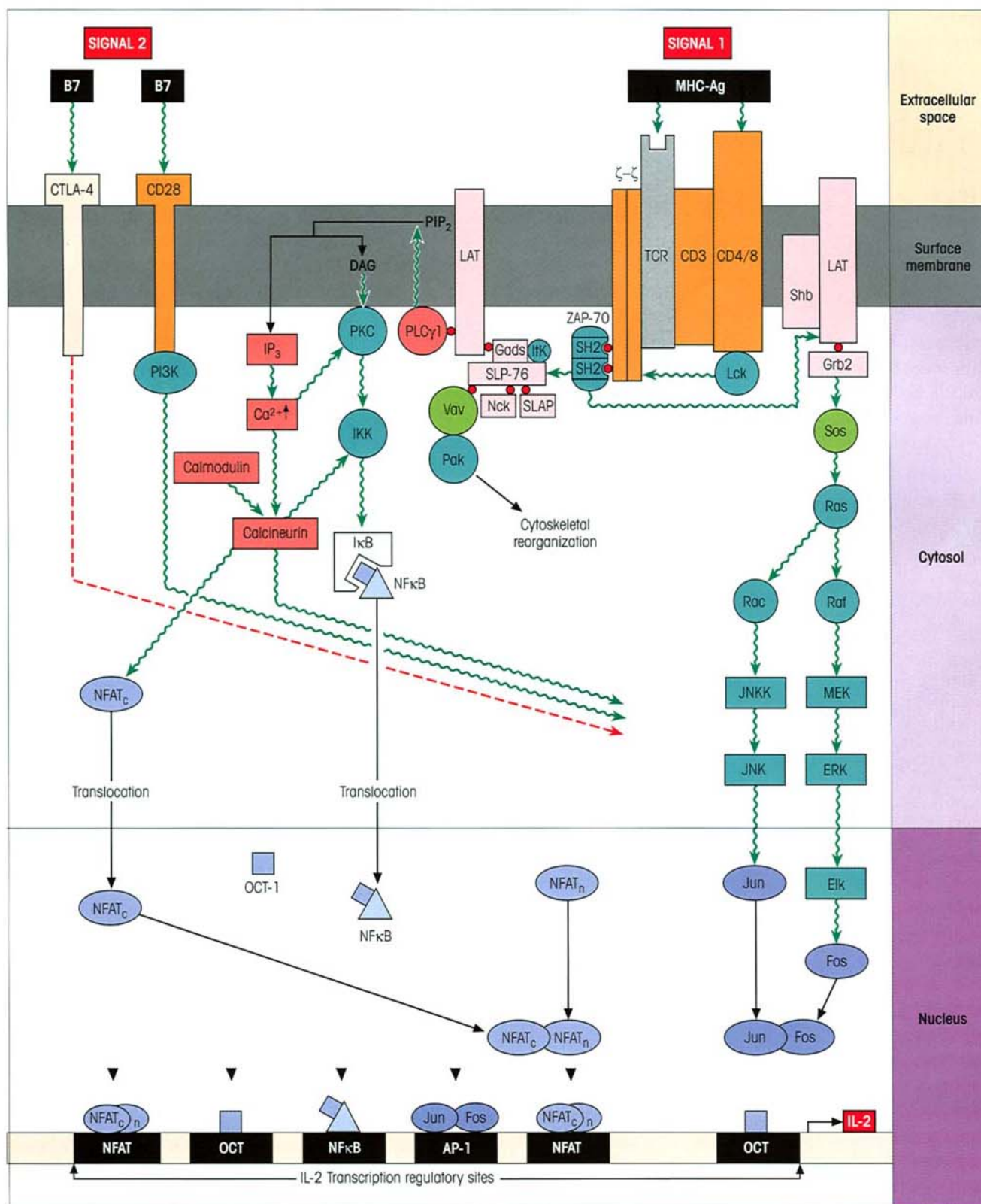
The phosphatidylinositol pathway

Within 15 seconds of TCR stimulation, the $\gamma 1$ isoform of **phospholipase C** (PLC $\gamma 1$), an enzyme which (like the $\gamma 2$ isoform) activates the phosphatidylinositol pathway, is phosphorylated and its catalytic activity increased. This early increase in phospholipase C activity accelerates the hydrolysis of phosphatidylinositol 4,5-bisphosphate (PIP₂) to diacylglycerol (DAG) and inositol 1,4,5-triphosphate (IP₃) (figure 9.7). The triphosphate binds to specific receptors on specialized calcium storage vesicles and triggers the release of Ca²⁺ into the cytosol; this is supplemented by an influx from the external milieu. The **raised Ca²⁺** level has at least two consequences. First, it synergizes with diacylglycerol to activate **protein kinase C (PKC)**; second, it acts together with **calmodulin** to increase the activity of **calcineurin**.

Control of IL-2 gene transcription

Transcription of IL-2 is one of the key elements in preventing the signaled T-cell from lapsing into anergy and is controlled by multiple receptors for transcriptional factors in the promoter region (figure 9.7). The MAP kinase JNK phosphorylates Jun, which then binds as a binary complex with Fos to the **AP-1** site, deletion of which abrogates 90% of IL-2 enhancer activity.

Under the influence of calcineurin, the cytoplasmic component of the nuclear factor of activated T-cells (NFAT_c) becomes activated and translocates to the nucleus where it forms a binary complex with NFAT_n, its partner which is constitutively expressed in the nucleus. The NFAT complex binds to two different IL-2 regulatory sites (figure 9.7). Note here that the calcineurin effect is blocked by the anti-T-cell drugs cyclosporin and FK506 (see Chapter 17). PKC- and calcineurin-dependent pathways synergize in activating the multisubunit I κ B kinase (IKK), which phosphorylates the inhibitor I κ B thereby targeting it for ubiquitination and subsequent degradation by the proteasome. Loss of I κ B from the I κ B–NF κ B complex exposes the nuclear localization signal on the NF κ B transcription factor which then swiftly enters the nucleus. In addition, the ubiquitous transcription



factor **Oct-1** interacts with specific octamer-binding sequence motifs.

We have concentrated on IL-2 transcription as an early and central consequence of T-cell activation, but more than 70 genes are newly expressed within 4 hours of T-cell activation, leading to proliferation and the synthesis of several cytokines and their receptors (see Chapter 10).

Further thoughts on T-cell triggering

A serial TCR engagement model for T-cell activation

We have already commented that the major docking forces which conjugate the APC and its T-lymphocyte counterpart must come from the complementary accessory molecules such as ICAM-1/LFA-1 and LFA-

3/CD2, rather than through the relatively low affinity TCR–MHC plus peptide links (figure 9.3). Nonetheless, cognate antigen recognition by the TCR remains a *sine qua non* for T-cell activation. Fine, but how can as few as 100 MHC–peptide complexes on an APC, through their low affinity complexing with TCRs, effect the Herculean task of sustaining a raised intracellular calcium flux for the 60 minutes required for full cell activation? Any fall in calcium flux, as may be occasioned by adding an antibody to the MHC, and NFAT_c dutifully returns from the nucleus to its cytoplasmic location, so aborting the activation process.

Surprisingly, Valitutti and Lanzavecchia have shown that as few as 100 MHC–peptide complexes on an APC can downregulate 18 000 TCRs on its cognate T-lymphocyte partner. They suggest that each MHC–peptide complex can *serially* engage up to 200 TCRs. In their model, conjugation of an MHC–peptide dimer with two TCRs (cf. p. 100) activates signal transduction, phosphorylation of the CD3-associated ζ chains with subsequent downstream events, and then down-regulation of those TCRs. Intermediate affinity binding favors dissociation of the MHC–peptide, freeing it to engage and trigger another TCR, so sustaining the required intracellular activation events. The model for **agonist** action would also explain why peptides giving interactions of lower or higher affinity than the optimum could behave as **antagonists** (figure 9.8). The important phenomenon of modified peptides behaving as **partial agonists**, with differential effects on the outcome of T-cell activation, is addressed in the legend to figure 9.8.

The immunological synapse

Experiments using peptide–MHC and ICAM-1 molecules labeled with different fluorochromes and inserted into a planar lipid bilayer on a glass support have provided evidence for the idea that T-cell activation occurs in the context of an **immunological synapse**. A clustered area of integrins acts as an anchor to permit optimal interaction between the opposing cell surfaces. Initially unstable TCR–MHC interactions occur in a broad outer ring surrounding the integrins. The peptide–MHC molecules then move towards the center of the synapse, changing places with the adhesion molecules which now form the outer ring (figure 9.9). It has been suggested that the generation of the immunological synapse only occurs after a certain initial threshold level of TCR triggering has been achieved, its formation being dependent upon cytoskeletal reorganization and leading to potentiation of the signal.

Figure 9.7. T-cell signaling leads to activation. The signals through the MHC–antigen complex (signal 1) and costimulator B7 (signal 2) initiate a protein kinase cascade and a rise in intracellular calcium, thereby activating transcription factors which control entry in the cell cycle from G0 and regulate the expression of IL-2 and many other cytokines. Itk is a member of the Tec family of kinases which associates with a multimeric complex containing a number of adaptor proteins involved in PLC γ 1 activation and thereby stimulates the IP₃ and DAG pathways. Upon activation by calcineurin, NFAT_c translocates from the cytoplasm to the nucleus and complexes with the nuclear component NFAT_n. The complex then binds to the NFAT transcription sites. The I κ B kinase (IKK) mediates the release of NF κ B from its inhibitor I κ B, whence it translocates to the nucleus and binds to a specific regulatory site. The Ras pathway activates the Raf, MEK, ERK and Elk kinases, finally activating the transcription factor Fos. Ras also activates Rac, which then phosphorylates JNKK, the kinase which activates JNK, and finally the transcription factor Jun which, complexed with Fos, binds to the AP-1 transcription site. B7 can incite a negative signal through CTLA-4. Note that the adaptor protein SLAP, also called Fyb, can play an inhibitory role in signaling by abrogating the function of SLP-76. The scheme presented omits several molecules which are thought to play important additional roles in signal transduction. As well as Lyk, other Src family PTKs, such as Fyn and Lyn, can phosphorylate ITAM motifs associated with a number of receptors including TCR, BCR and FcR. Transmembrane adaptors not shown include SIT (SHP-2 interacting transmembrane adaptor) and TRIM (T-cell receptor interacting molecule): the latter may act through the Grb2 adaptor as a positive regulator, whereas SIT may be a negative regulator. Abbreviations: DAG, diacylglycerol; ERK, extracellular signal regulated kinase; IP₃, inositol triphosphate; JNK, Jun N-terminal kinase; LAT, linker for activated T-cells; NF κ B, nuclear factor κ B; NFAT, nuclear factor of activated T-cells; OCT-1, octamer-binding factor; Pak, p21-activated kinase; PI3K, phosphatidylinositol 3-kinase; PIP₂, phosphatidylinositol diphosphate; PKC, protein kinase C; PLC, phospholipase C; SH2, Src-homology domain 2; SLAP, SLP-76-associated phosphoprotein; SLP-76, SH2-domain containing leukocyte-specific 76 kDa phosphoprotein; ZAP-70, ζ chain-associated protein kinase. , Positive signal transduction; , negative signal transduction; , adaptor proteins; , guanine nucleotide exchange factors; , kinases; , transcription factors; , other molecules.

T-cell mutants lacking CD45 do not possess signal transduction capacity was at first sight deemed to be strange because CD45 has phosphatase activity and was thought thereby to downregulate signaling. However, the Lck kinase in the CD45-deficient cells is phosphorylated on tyrosine-505 which is a negative regulatory site for kinase activity; hence dephosphorylation by CD45 activates the Lck enzyme and the paradox is resolved.

B-CELLS RESPOND TO THREE DIFFERENT TYPES OF ANTIGEN

1 Type 1 thymus-independent antigens

Certain antigens, such as bacterial lipopolysaccharides, at a high enough concentration, have the ability to activate a substantial proportion of the B-cell pool polyclonally, i.e. without reference to the antigen specificity of the surface receptor hypervariable regions. They do this through binding to a surface molecule which bypasses the early part of the biochemical pathway mediated by the specific antigen receptor. At concentrations which are too low to cause polyclonal activation through unaided binding to these mitogenic bypass molecules, the B-cell population with Ig receptors specific for these antigens will selectively and passively focus them on their surface, where the resulting high local concentration will suffice to drive the activation process (figure 9.10a).

2 Type 2 thymus-independent antigens

Certain linear antigens which are not readily degraded in the body and which have an appropriately spaced, highly repeating determinant—*Pneumococcus* polysaccharide, ficoll, D-amino acid polymers and poly-

vinylpyrrolidone, for example—are also thymus-independent in their ability to stimulate B-cells directly without the need for T-cell involvement. They persist for long periods on the surface of specialized macrophages located at the subcapsular sinus of the lymph nodes and the splenic marginal zone, and can bind to antigen-specific B-cells with great avidity through their multivalent attachment to the complementary Ig receptors which they cross-link (figure 9.10b).

In general, the thymus-independent antigens give rise to predominantly low affinity IgM responses, some IgG3 in the mouse, and relatively poor, if any, memory. Neonatal B-cells do not respond well to type 2 antigens and this has important consequences for the efficacy of carbohydrate vaccines in young children.

3 Thymus-dependent antigens

The need for collaboration with T-helper cells

Many antigens are thymus-dependent in that they provoke little or no antibody response in animals which have been thymectomized at birth and have few T-cells (Milestone 9.1). Such antigens cannot fulfil the molecular requirements for direct stimulation; they may be univalent with respect to the specificity of each determinant; they may be readily degraded by phagocytic cells; and they may lack mitogenicity. If they bind to B-cell receptors, they will sit on the surface just like a hapten and do nothing to trigger the B-cell (figure 9.11). Cast your mind back to the definition of a hapten—a small molecule like dinitrophenyl (DNP) which binds to preformed antibody (e.g. the surface receptor of a specific B-cell) but fails to stimulate antibody production (i.e. stimulate the B-cell). Remember also that haptens become immunogenic when coupled to an

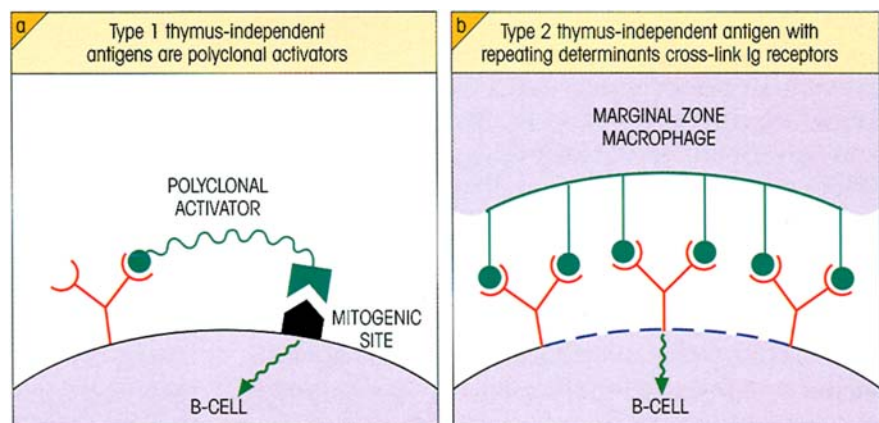


Figure 9.10. B-cell recognition of (a) type 1 and (b) type 2 thymus-independent antigens. The complex gives a sustained signal to the B-cell because of the long half-life of this type of molecule. —, activation signal; —, surface Ig receptor; — — —, cross-linking of receptors.

Milestone 9.1 —T–B Collaboration for Antibody Production

In the 1960s, as the mysteries of the thymus were slowly unraveled, our erstwhile colleagues pushing back the frontiers of knowledge discovered that neonatal thymectomy in the mouse abrogated not only the cellular rejection of skin grafts, but also the antibody response to some but not all antigens (figure M9.1.1). Subsequent investigations showed that both thymocytes and bone marrow cells were needed for optimal antibody responses to such **thymus-dependent antigens** (figure M9.1.2). By carrying out these transfers with cells from animals bearing a recognizable chromosome marker (T6), it became evident that the antibody-forming cells were derived from the bone marrow inoculum, hence the nomenclature ‘T’ for Thymus-derived lymphocytes and ‘B’ for antibody-forming cell precursors originating in the Bone marrow. This convenient nomenclature has stuck even though bone marrow contains embryonic T-cell precursors since the immunocompetent T- and B-cells differentiate in the thymus and bone marrow respectively (see Chapter 12).

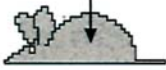
THYMECTOMY	Ag INJECTED	ANTIBODY RESPONSE
Sham	TETANUS TOXOID or PNEUMOCOCCAL POLYSACCHARIDE S III 	+++
Neonatal	TETANUS TOXOID 	-
Neonatal	PNEUMOCOCCAL POLYSACCHARIDE 	+++

Figure M9.1.1. The antibody response to some antigens is thymus-dependent and, to others, thymus-independent. The response to tetanus toxoid in neonatally thymectomized animals could be restored by the injection of thymocytes.

Cells injected	None	Thymocytes (T)	Bone marrow (B)	Thymocytes & Bone marrow
X-irradiated recipient inj. with thymus-dependent Ag				
Production of Ab	-	-	+	+++

Figure M9.1.2. The antibody response to a thymus-dependent antigen requires two different cell populations. Different populations of cells from a normal mouse histocompatible with the recipient (i.e. of the same H-2 haplotype) were injected into recipients which had been X-irradiated to destroy their own lymphocyte responses. They were then primed with a thymus-dependent antigen such as sheep red blood cells (i.e. an antigen

which fails to give a response in neonatally thymectomized mice; figure M9.1.1) and examined for the production of antibody after 2 weeks. The small amount of antibody synthesized by animals receiving bone marrow alone is due to the presence of thymocyte precursors in the cell inoculum which differentiate in the intact thymus gland of the recipient.

appropriate carrier protein (see p. 80). Building on the knowledge that both T- and B-cells are necessary for antibody responses to thymus-dependent antigens (Milestone 9.1), we now know that the carrier functions to stimulate T-helper cells which cooperate with B-cells to enable them to respond to the hapten by providing accessory signals (figure 9.11). It should also be evident from figure 9.11 that, while one determinant on a typical protein antigen is behaving as a hapten in binding to the B-cell, the other determinants subserve a carrier function in recruiting T-helper cells.

Antigen processing by B-cells

The need for **physical linkage of hapten and carrier** strongly suggests that T-helpers must recognize the carrier determinants on the responding B-cell in order to provide the relevant accessory stimulatory signals. However, since T-cells only recognize processed membrane-bound antigen in association with MHC molecules, the T-helpers cannot recognize native antigen bound simply to the Ig receptors of the B-cell as naively depicted in figure 9.11. All is not lost, however,

since **primed B-cells can present antigen to T-helper cells**—in fact, they work at much lower antigen concentrations than conventional presenting cells because they can focus antigen through their surface receptors. Antigen bound to surface Ig is internalized in endosomes which then fuse with vesicles containing MHC class II molecules with their invariant chain. Processing of the protein antigen then occurs as described in Chapter 5 (see figure 5.18) and the resulting antigenic peptide is recycled to the surface in association with the class II molecules where it is available for recognition by specific T-helpers (figures 9.12 and 9.13). The need for the physical union of hapten and carrier is now revealed; the hapten leads the carrier to be processed into the cell which is programmed to make

antihapten antibody and, following stimulus by the T-helper-recognizing processed carrier, it will carry out its program and ultimately produce antibodies which react with the hapten (is there no end to the wiliness of nature?!).

THE NATURE OF B-CELL ACTIVATION

B-cells are stimulated by cross-linking surface Ig

Cross-linking of the B-cell receptor (BCR), for example by anti-IgM conjugated to insoluble Sepharose particles or by polymeric type 2 thymus-independent antigens with repeating determinants, induces the early activation events. Coligation of the pan-B-cell marker CD19 with surface Ig (sIg) reduces the threshold for cell activation 100-fold. *In vivo* this could be brought about by bridging the Ig and CR2 complement receptors on the B-cell surface by antigen–C3d complexes bound to the surface of APCs, since CD19 and CR2 (CD21) molecules enjoy mutual association.

Within 1 minute of surface Ig ligation, Src family kinases rapidly phosphorylate the Syk kinase, Bruton's tyrosine kinase (Btk) and the ITAMs on the sIg receptor chains Ig- α and Ig- β . Syk binds to the Ig- α and Ig- β ITAMs, analogously to ZAP-70 binding to the ITAMs on the TCR-associated ζ chains. CD19 also becomes phosphorylated, possibly by Lyn, thereby creating binding sites for the SH2 domains of phosphatidylinositol 3-kinase and the guanine nucleotide exchange factor Vav. Although B-cells lack the LAT and SLP-76 adaptors of T-cells (cf. figure 9.7), they possess a homolog of SLP-76 variously referred to as SLP-65, BLNK

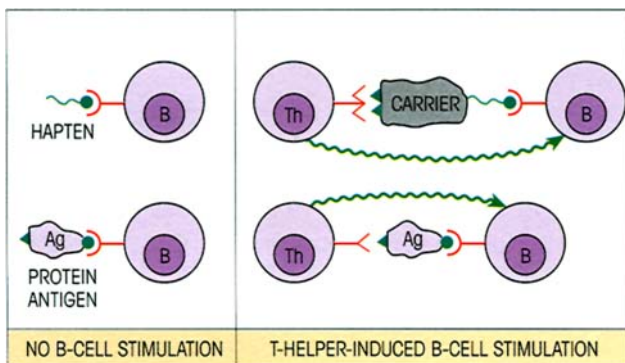
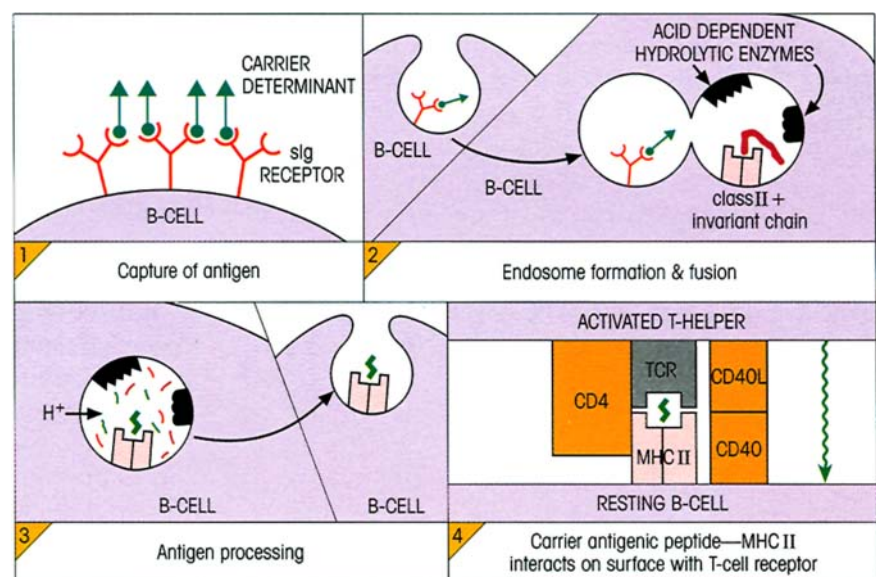


Figure 9.11. T-helper cells cooperate through protein carrier determinants to help B-cells respond to hapten or equivalent determinants on antigens by providing accessory signals. (For simplicity we are ignoring the MHC component and epitope processing in T-cell recognition, but we won't forget it.)

Figure 9.12. B-cell handling of a thymus-dependent antigen. Antigen captured by the surface Ig receptor is internalized within an endosome, processed and expressed on the surface with MHC class II (cf. figure 5.18). Costimulatory signals through the CD40–CD40L (CD154) interaction are required for the activation of the resting cell by the T-helper. sIg cross-linking by antigen on the surface of an antigen-presenting cell is likely during secondary responses within the germinal centers when complement-containing complexes on follicular dendritic cells interact with B-lymphoblasts. , activation signal; , cross-linking of receptors.



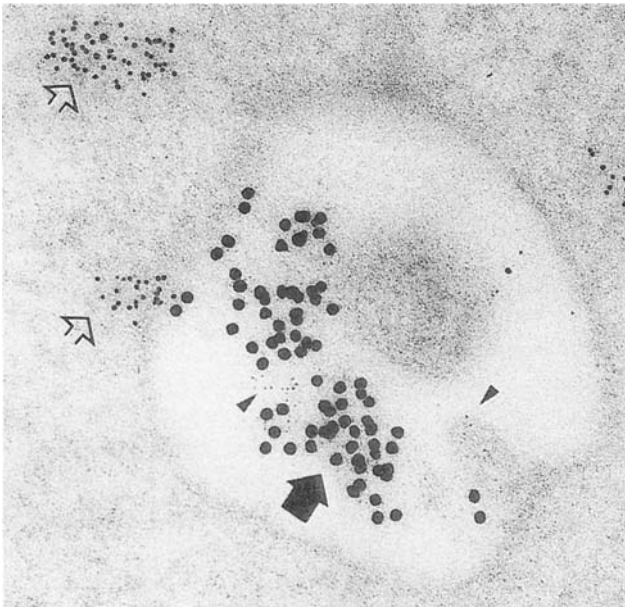


Figure 9.13. Demonstration that endocytosed B-cell surface Ig receptors enter cytoplasmic vesicles geared for antigen processing. Surface IgG was cross-linked with goat anti-human Ig and rabbit anti-goat Ig conjugated to 15-nm gold beads (large, dark arrow). After 2 minutes, the cell sections were prepared and stained with anti-HLA-DR invariant chain (2nm gold; arrowheads) and an antibody to a cathepsin protease (5nm gold; open arrows). Thus the internalized IgG is exposed to proteolysis in a vesicle containing class II molecules. The presence of invariant chain shows that the class II molecules derive from the endoplasmic reticulum and Golgi, not from the cell surface. Note the clever use of different-sized gold particles to distinguish the antibodies used for localizing the various intravesicular proteins, etc. (Photograph reproduced with permission from the authors and the publishers from Guagliardi L.E. *et al.* (1990) *Nature* 343, 133. Copyright © 1990 Macmillan Magazines Ltd.)

(B-cell linker protein) or BASH (B-cell adaptor containing SH2 domain). SLP-65 is phosphorylated by Syk and forms a multimeric complex incorporating Vav, Btk and phospholipase $C\gamma 2$ (PLC $\gamma 2$), the latter kicking off the PIP $_2$ pathway. Just as in the T-cell, this generates inositol triphosphate and diacylglycerol with a subsequent rise in intracellular calcium and activation of protein kinase C (cf. figure 9.7). The enhancer-binding protein NF κ B is released from its inhibitor I κ B on activation of protein kinase C and moves into the nucleus, where its appearance is associated with κ light chain gene transcription. The SLP-65 adaptor also links B-cell signaling into activation of the ERK and JNK pathways.

The BCR cross-linking model seems appropriate for an understanding of stimulation by type 2 thymus-independent antigens, since their repeating determinants ensure strong binding to, and cross-linking of, multiple Ig receptors on the B-cell surface to form aggregates which persist owing to the long half-life of the antigen and sustain the high intracellular calcium needed for activation. On the other hand, type 1 T-independent antigens, like the T-cell polyclonal activators, probably bypass the specific receptor and act directly on downstream molecules such as diacylglycerol and protein kinase C since Ig- α and Ig- β are not phosphorylated.

T-helper cells activate resting B-cells

T-dependent antigens pose a different problem, since they are usually univalent with respect to B-cell receptors, i.e. each epitope appears once on a monomeric protein, and as a result they cannot cross-link the surface Ig. However, we have discussed in some detail how antigen captured by a B-cell receptor can be internalized and processed for surface presentation as a peptide complexed with class II MHC (cf. figure 9.12). This can now be recognized by the TCR of a carrier-specific T-helper cell and, with the assistance of costimulatory signals arising from the interaction of **CD40 with its ligand CD40L** (cf. figure 9.12), B-cell activation is ensured.

In effect, the B-lymphocyte is acting as an antigen-presenting cell and, as mentioned above, it is very efficient because of its ability to concentrate the antigen by focusing onto its surface Ig. Nonetheless, although a preactivated T-helper can mutually interact with and stimulate a resting B-cell, a *resting* T-cell can only be triggered by a B-cell that has acquired the B7 costimulator and this is only present on activated, not resting, B-cells.

Presumably the immune complexes on follicular dendritic cells in germinal centers of secondary follicles can be taken up by the B-cells for presentation to T-helpers, but, additionally, the complexes could cross-link the sIg of the B-cell blasts and drive their proliferation in a T-independent manner. This would be enhanced by the presence of C3 in the complexes since the B-cell complement receptor (CR2) is comitogenic.

SUMMARY

Immunocompetent T- and B-cells differ in many respects

- The antigen-specific receptors, TCR/CD3 on T-cells and surface Ig on B-cells, provide a clear distinction between these two cell types.
- T- and B-cells differ in their receptors for C3d, IgG and certain viruses.
- There are distinct polyclonal activators of T-cells (PHA, anti-CD3) and of B-cells (anti-Ig, Epstein–Barr virus).

T-lymphocytes and antigen-presenting cells interact through pairs of accessory molecules

- The docking of T-cells and APCs depends upon strong mutual interactions between complementary molecular pairs on their surfaces: MHC II–CD4, MHC I–CD8, VCAM-1–VLA-4, ICAM-1–LFA-1, LFA-3–CD2, B7–CD28 (and CTLA-4).

Activation of T-cells requires two signals

- Two signals activate T-cells, but one alone produces unresponsiveness (anergy).
- One signal is provided by the low affinity cognate TCR–MHC plus peptide interaction.
- The second costimulatory signal is mediated through ligation of CD28 by B7.

Protein tyrosine phosphorylation is an early event in T-cell signaling

- The TCR signal is transduced and amplified through a protein tyrosine kinase (PTK) enzymic cascade.
- CD4/TCR colocation leads to phosphorylation of ITAM sequences on CD3-associated ζ chains by the CD4-associated Lck PTK. The phosphorylated ITAMs bind and then activate the ZAP-70 kinase.

Downstream events following TCR signaling

- Nonenzymic adaptor proteins form multimeric complexes with kinases and guanine nucleotide exchange factors (GEFs).
- Hydrolysis of phosphatidylinositol diphosphate by phospholipase $C\gamma 1$ or $C\gamma 2$ produces inositol triphosphate (IP_3) and diacylglycerol (DAG).
- IP_3 mobilizes intracellular calcium.
- DAG and increased calcium activate protein kinase C.
- The raised calcium together with calmodulin also stimulates calcineurin activity.
- Activation of Ras by the guanine nucleotide exchange factor Sos sets off a kinase cascade operating through Raf,

the MAP kinase kinase MEK and the MAP kinase ERK. CD28 through PI3 kinase can also influence MAP kinase.

- The transcription factors Fos and Jun, NFAT and NF κ B are activated by MAP kinase, calcineurin and PKC, respectively, and bind to regulatory sites in the IL-2 promoter region.
- A small number of MHC–peptide complexes can serially trigger a much larger number of TCRs thereby providing the sustained signal required for activation.
- Initial binding of integrins facilitates the formation of an immunological synapse, the core of which exchanges integrins for TCR interacting with MHC–peptide.
- B7 delivers a negative signal through CTLA-4, and Cbl family adaptor molecules are also involved in negative signaling pathways.
- The phosphatase domains on CD45 are required to remove phosphates at inhibitory sites on kinases.

B-cells respond to three different types of antigen

- Type 1 thymus-independent antigens are polyclonal activators focused onto the specific B-cells by sIg receptors.
- Type 2 thymus-independent antigens are polymeric molecules which cross-link many sIg receptors and, because of their long half-lives, provide a persistent signal to the B-cell.
- Thymus-dependent antigens require the cooperation of helper T-cells to stimulate antibody production by B-cells.
- Antigen captured by specific sIg receptors is taken into the B-cell, processed and expressed on the surface as a peptide in association with MHC II.
- This complex is recognized by the T-helper cell which activates the resting B-cell.
- The ability of protein carriers to enable the antibody response to haptens is explained by T–B collaboration, with T-cells recognizing the carrier and B-cells the hapten.

The nature of B-cell activation

- Cross-linking of surface Ig receptors (e.g. by type 2 thymus-independent antigens) activates B-cells.
- T-helper cells activate resting B-cells through TCR recognition of MHC II–carrier peptide complexes and co-stimulation through CD40L–CD40 interactions (analogous to the B7–CD28 second signal for T-cell activation).

See the accompanying website (www.roitt.com) for multiple choice questions.

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INTRODUCTION

We have dwelt upon the early events in lymphocyte activation consequent upon the engagement of the T-cell receptor (TCR) and the provision of an appropriate costimulatory signal. A complex series of tyrosine and serine/threonine phosphorylation reactions produces the factors which push the cell into the mitotic cycle and drive clonal proliferation and the differentiation to effectors. A succession of genes are upregulated by T-cell activation. Within the first half hour, nuclear transcription factors such as Fos/Jun and NFAT, which regulate interleukin-2 (IL-2) expression and the cellular protooncogene *c-myc*, are expressed, but the next few hours see the synthesis of a range of **soluble cytokines and their receptors**. Much later we see molecules like the transferrin receptor related to cell division and very late antigens such as the adhesion molecule VLA-1.

CYTOKINES ACT AS INTERCELLULAR MESSENGERS

In contrast with the initial activation of T-cells and T-dependent B-cells, which involves intimate contact

with the antigen-presenting cells (APCs), subsequent proliferation and maturation of the response are orchestrated by soluble mediators generically termed cytokines (figure 10.1). With the realization that a given cytokine can be produced by many different cell types, terms such as lymphokine or monokine have become somewhat obsolete (table 10.1).

Cytokine action is transient and usually short range

These low molecular weight secreted proteins, usually 15–25 kDa, mediate cell growth, inflammation, immunity, differentiation, migration and repair. Because they regulate the amplitude and duration of the immune-inflammatory responses, they must be produced in a transient manner tightly regulated by the presence of foreign material, and it is relevant that the AU-rich sequences in the 3' untranslated regions of the mRNA of many cytokines are correlated directly with rapid degradation and therefore short half-life. Unlike endocrine hormones, the majority of cytokines normally act locally in a paracrine or even autocrine fashion. Thus cytokines derived from lymphocytes rarely persist in the circulation, but nonlymphoid cells can be triggered by bacterial products to release cytokines which may be detected in the bloodstream, often to the detriment of the host. Certain cytokines,

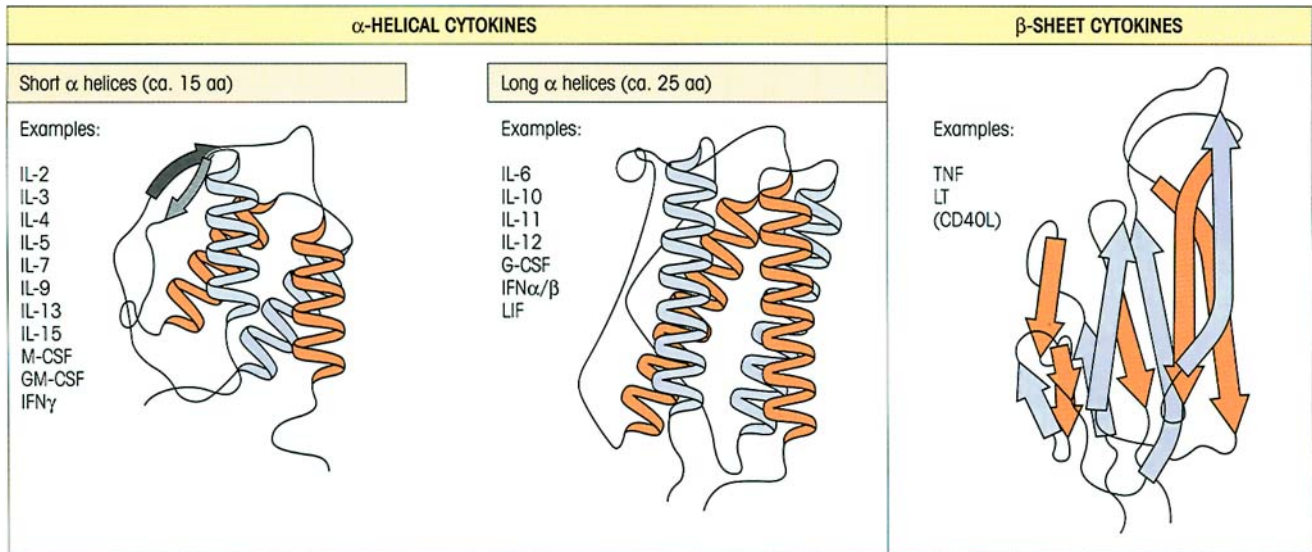


Figure 10.1. Cytokine structures. Cytokines can be divided into a number of different structural groups. Illustrated here are three of the main types of structure and some named examples of each type: (a) four short (~15 amino acids) α -helices, (b) four long (~25 amino acids) α -helices and (c) a β -sheet structure. (Reproduced with permission from Michal G. (ed.) (1999) *Biochemical Pathways: An Atlas of Biochemistry and Molecular Biology*. John Wiley & Sons, New York.)

including IL-1 and tumor necrosis factor (TNF), also exist in membrane forms which can exert their stimulatory effects without becoming soluble.

Cytokines act through cell surface receptors

Cytokines are highly potent, often acting at femtomolar (10^{-15} M) concentrations, combining with small numbers of high affinity cell surface receptors to produce changes in the pattern of RNA and protein synthesis. A common feature in the triggering of the cytokine receptors is the ligand-induced association of receptor subunits which allows signal transduction through the interplay of their juxtaposed cytoplasmic domains. There are six major cytokine receptor structural families (figure 10.2).

Hematopoietin receptors

These are the largest family, sometimes referred to simply as the cytokine receptor superfamily. These receptors generally consist of one or two polypeptide chains responsible for cytokine binding and an additional shared (common or 'c') chain involved in signal transduction. The γ c (CD132) chain is used by the IL-2

receptor (figure 10.2a) and IL-4, IL-7, IL-9, IL-13 and IL-15 receptors, a β c (CDw131) chain by IL-3, IL-5 and granulocyte-macrophage colony-stimulating factor (GM-CSF) receptors, and gp130 (CD130) shared chain by the IL-6, IL-11, IL-12, oncostatin M, ciliary neurotrophic factor and leukemia inhibitory factor (LIF) receptors.

Interferon receptors

These also consist of two polypeptide chains and, in addition to the IFN α , IFN β and IFN γ receptors (figure 10.2b), this family includes the IL-10 receptor.

TNF receptors

The receptor polypeptides are cysteine rich and trimerize following cytokine binding. They include the tumor necrosis factor (TNF) receptor (figure 10.2c), lymphotoxin (LT) and nerve growth factor (NGF) receptors, and the cell surface-associated molecules CD40 and CD95 (Fas).

IgSF cytokine receptors

Immunoglobulin superfamily members are broadly utilized in many aspects of cell biology (cf. p. 245) and include the IL-1 receptor (figure 10.2d), and the macrophage colony-stimulating factor (M-CSF) and stem cell factor (SCF/c-kit) receptors.

Chemokine receptors

As we shall discover shortly, these comprise a family of approximately 20 different G-protein-coupled,

Table 10.1. Cytokines: their origin and function.

CYTOKINE	SOURCE	EFFECTOR FUNCTION
INTERLEUKINS		
IL-1 α , IL-1 β	Mono, M ϕ , DC, NK, B, Endo	Costimulates T activation by enhancing production of cytokines including IL-2 and its receptor; enhances B proliferation and maturation; NK cytotoxicity; induces IL-1, -6, -8, TNF, GM-CSF and PGE ₂ by M ϕ ; proinflammatory by inducing chemokines and ICAM-1 and VCAM-1 on endothelium; induces fever, APP, bone resorption by osteoclasts
IL-2	Th1	Induces proliferation of activated T- and B-cells; enhances NK cytotoxicity and killing of tumor cells and bacteria by monocytes and M ϕ
IL-3	T, NK, MC	Growth and differentiation of hematopoietic precursors; MC growth
IL-4	Th2, Tc2, NK, NK-T, $\gamma\delta$ T, MC	Induces Th2 cells; stimulates proliferation of activated B, T, MC; upregulates MHC class II on B and M ϕ , and CD23 on B; downregulates IL-12 production and thereby inhibits Th1 differentiation; increases M ϕ phagocytosis; induces switch to IgG1 and IgE
IL-5	Th2, MC	Induces proliferation of eosino and activated B; induces switch to IgA
IL-6	Th2, Mono, M ϕ , DC, BM stroma	Differentiation of myeloid stem cells and of B into plasma cells; induces APP; enhances T proliferation
IL-7	BM and thymic stroma	Induces differentiation of lymphoid stem cells into progenitor T and B; activates mature T
IL-8	Mono, M ϕ , Endo	Mediates chemotaxis and activation of neutrophils
IL-9	Th	Induces proliferation of thymocytes; enhances MC growth; synergizes with IL-4 in switch to IgG1 and IgE
IL-10	Th (Th2 in mouse), Tc, B, Mono, M ϕ	Inhibits IFN γ secretion by mouse, and IL-2 by human, Th1 cells; downregulates MHC class II and cytokine (including IL-12) production by mono, M ϕ and DC, thereby inhibiting Th1 differentiation; inhibits T proliferation; enhances B differentiation
IL-11	BM stroma	Promotes differentiation of pro-B and megakaryocytes; induces APP
IL-12	Mono, M ϕ , DC, B	Critical cytokine for Th1 differentiation; induces proliferation and IFN γ production by Th1, CD8 ⁺ and $\gamma\delta$ T and NK; enhances NK and CD8 ⁺ T cytotoxicity
IL-13	Th2, MC	Inhibits activation and cytokine secretion by M ϕ ; co-activates B proliferation; upregulates MHC class II and CD23 on B and mono; induces switch to IgG1 and IgE; induces VCAM-1 on endo
IL-15	T, NK, Mono, M ϕ , DC, B	Induces proliferation of T-, NK and activated B and cytokine production and cytotoxicity in NK and CD8 ⁺ T; chemotactic for T; stimulates growth of intestinal epithelium
IL-16	Th, Tc	Chemoattractant for CD4 T, mono and eosino; induces MHC class II
IL-17	T	Proinflammatory; stimulates production of cytokines including TNF, IL-1 β , -6, -8, G-CSF
IL-18	M ϕ , DC	Induces IFN γ production by T; enhances NK cytotoxicity
IL-19	Mono	Modulation of Th1 activity
IL-20	Keratinocytes?	Regulation of inflammatory responses to skin?
IL-21	Th	Regulation of hematopoiesis; NK differentiation; B activation; T costimulation
IL-22	T	Inhibits IL-4 production by Th2
IL-23	DC	Induces proliferation and IFN γ production by Th1; induces proliferation of memory cells
COLONY STIMULATING FACTORS		
GM-CSF	Th, M ϕ , Fibro, MC, Endo	Stimulates growth of progenitors of mono, neutro, eosino and baso; activates M ϕ
G-CSF	Fibro, Endo	Stimulates growth of neutro progenitors
M-CSF	Fibro, Endo, Epith	Stimulates growth of mono progenitors
SLF	BM stroma	Stimulates stem cell division (<i>c-kit</i> ligand)
TUMOR NECROSIS FACTORS		
TNF (TNF α)	Th, Mono, M ϕ , DC, MC, NK, B	Tumor cytotoxicity; cachexia (weight loss); induces cytokine secretion; induces E-selectin on endo; activates M ϕ ; antiviral
Lymphotoxin (TNF β)	Th1 Tc	Tumor cytotoxicity; enhances phagocytosis by neutro and M ϕ ; involved in lymphoid organ development; antiviral
INTERFERONS		
IFN α	Leukocytes	Inhibits viral replication; enhances MHC class I
IFN β	Fibroblasts	Inhibits viral replication; enhances MHC class I
IFN γ	Th1, Tc1, NK	Inhibits viral replication; Enhances MHC class I and II; activates M ϕ ; induces switch to IgG2a; antagonizes several IL-4 actions; inhibits proliferation of Th2
OTHERS		
TGF β	Th3, B, M ϕ , MC	Proinflammatory by, e.g., chemoattraction of mono and M ϕ but also anti-inflammatory by, e.g. inhibiting lymphocyte proliferation; induces switch to IgA; promotes tissue repair
LIF	Thymic epith, BM stroma	Induces APP
Eta-1	T	Stimulates IL-12 production and inhibits IL-10 production by M ϕ
Oncostatin M	T, M ϕ	Induces APP

APP, acute phase proteins; B, B-cell; baso, basophil; BM, bone marrow; Endo, endothelium; eosino, eosinophil; Epith, epithelium; Fibro, fibroblast; GM-CSF, granulocyte-macrophage colony-stimulating factor; IL, interleukin; LIF, leukemia inhibitory factor; M ϕ , macrophage; MC, mast cell; Mono, monocyte; neutro, neutrophil; NK, natural killer; SLF, steel locus factor; T, T-cell; TGF β , transforming growth factor- β . Note that there is not an interleukin-14. This designation was given to an activity that, upon further investigation, could not be unambiguously assigned to a single cytokine. IL-8 is a member of the chemokine family. These cytokines are listed separately in table 10.3.

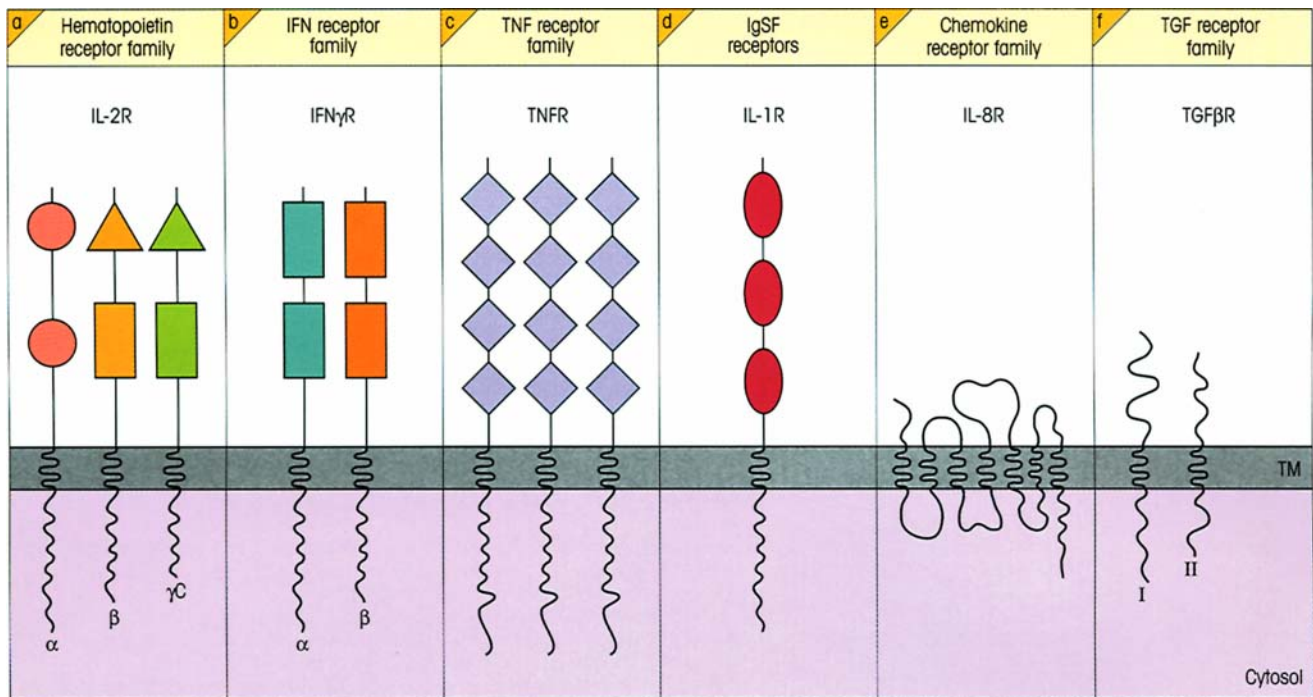


Figure 10.2. Cytokine receptor families. One example is shown for each family. (a) The **hematopoietin receptors** operate through a common subunit (γ c, β c or gp130, depending on the subfamily) which transduces the signal to the interior of the cell. In essence, binding of the cytokine to its receptor must initiate the signaling process by mediating hetero- or homodimer formation involving the common subunit. In some cases the cytokine is active when bound to the receptor either in soluble or membrane-bound form (e.g. IL-6). The IL-2 receptor is interesting with respect to its ligand binding. The α chain (CD25, reacting with the Tac monoclonal) of the receptor possesses two complement control protein structural domains and binds IL-2 with a low affinity; the β chain (CD122) has a membrane proximal fibronectin type III structural domain and a membrane distal cytokine receptor structural domain, and associates with the common γ chain (CD132) which has a similar structural organization. The β chain binds IL-2 with intermediate affinity. IL-2 binds to and dissociates from the α chain very rapidly but the same processes involving the β chain occur at two or three orders of magnitude more slowly. When the α , β and γ chains form a single recep-

tor, the α chain binds the IL-2 rapidly and facilitates its binding to a separate site on the β chain from which it can only dissociate slowly. Since the final affinity (K_d) is based on the ratio of dissociation to association rate constants, then $K_d = 10^{-4} \text{sec}^{-1} / 10^7 \text{M}^{-1} \text{sec}^{-1} = 10^{-11} \text{M}$, which is a very high affinity. The γ chain does not itself bind IL-2 but contributes towards signal transduction. (b) The **interferon receptor** family consists of heterodimeric molecules each of which bears two fibronectin type III domains. (c) The receptors for TNF and related molecules consist of a single polypeptide with four TNFR domains. The receptor trimerizes upon ligand binding and, in common with a number of other receptors, is also found in a soluble form which, when released from a cell following activation, can act as an antagonist. (d) Another group of receptors contains varying numbers of **Ig superfamily domains**, whereas (e) **chemokine receptors** are members of the G-protein-coupled receptor superfamily and have seven hydrophobic transmembrane domains. (f) The final family illustrated are the **TGF receptors** which require association between two molecules, referred to as TGF β type I and TGF β type II, for signaling to occur.

seven transmembrane segment, receptors (figure 10.2e).

TGF receptors

Receptors for transforming growth factors such as the TGF β receptor (figure 10.2f) possess cytoplasmic portions with serine/threonine kinase activity.

Signal transduction through cytokine receptors

The ligand-induced homo- or heterodimerization of cytokine receptor subunits represents a common theme for signaling by cytokines. The two major routes that are utilized are the Janus kinase (JAK)–STAT

and the Ras–MAP kinase pathways. Members of the cytokine receptor superfamily (hematopoietin receptors) lack catalytic domains and therefore associate with JAKs including JAK1, JAK2, JAK3 and Tyk2. Upon cytokine-induced receptor dimerization, the JAKs reciprocally phosphorylate, and thereby activate, each other. Once activated, they can phosphorylate the cytoplasmic domains of the receptors to create binding sites for SH2-containing signaling proteins. These include one or more of the group of transcription factors termed **STATs** (signal transducers and activators of transcription), which also become phosphorylated by the workaholic JAKs, an event which causes the STATs to dissociate from the receptor and dimerize. The dimerized STATs then translocate to the nucleus

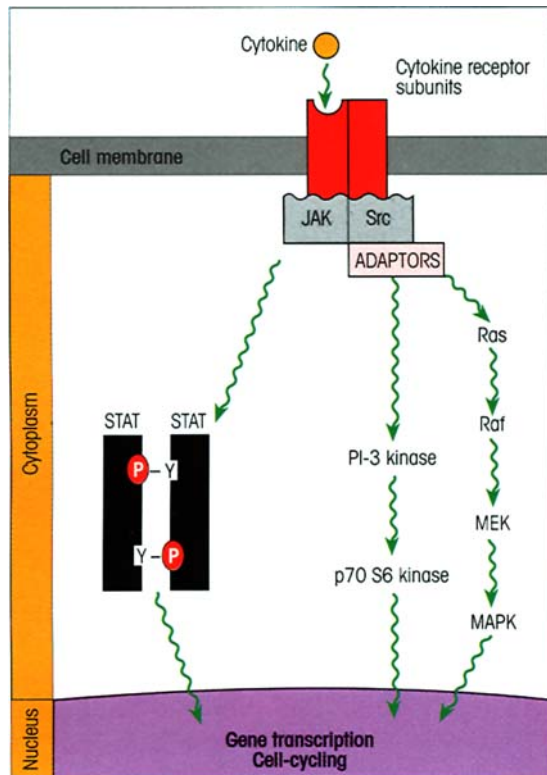


Figure 10.3. Cytokine receptor-mediated pathways for gene transcription. Cytokine-induced receptor oligomerization activates the associated JAK kinases. These phosphorylate the STAT DNA-binding transcription factors which dimerize and translocate to the nucleus. Cytokine receptors can also activate src family kinases which, via adaptor proteins such as Shc, Grb2 and SHP-2, could generate transcription factors via the two routes shown. MAPK, mitogen-activated protein kinases (see figure 9.7); MEK, kinase activator of MAPK; PI-3 kinase, phosphatidylinositol 3-kinase.

where they play an important role in pushing the cell through the mitotic cycle (figure 10.3). JAKs may also act through src family kinases to generate other transcription factors via the Ras–MAP kinase route (figure 10.3). Some cytokines also activate phosphatidylinositol 3-kinase (PI3K) and phospholipase C (PLC γ).

Downregulation of cytokine receptor-mediated signals involves members of the SOC (suppressor of cytokine signaling) and CIS (cytokine-inducible src homology domain 2 [SH2]-containing) families. Although their mode of action is still being evaluated, some of these molecules may function by directly blocking phosphorylation motifs on the receptor, or possibly by acting as scavengers of tyrosine phosphorylated proteins, targeting them for ubiquitin/proteasome (cf. p. 94) degradation.

Cytokines often have multiple effects

In general, cytokines are **pleiotropic**, i.e. exhibit multiple effects on a variety of cell types (table 10.1),

and there is considerable overlap and *redundancy* between them with respect to individual functions, partially accounted for by the sharing of receptor components and the utilization of common transcription factors. For example, many of the biological activities of IL-4 overlap with those of IL-13. However, it should be pointed out that virtually all cytokines have at least some unique properties.

Their roles in the generation of T- and B-cell effectors, and in the regulation of chronic inflammatory reactions (figure 10.4a and b), will be discussed at length later in this chapter. We should note here the important role of cytokines in the control of hematopoiesis (figure 10.4c). The differentiation of stem cells to become the formed elements of blood within the environment of the bone marrow is carefully nurtured through the production of cytokines by the stromal cells. These include GM-CSF, G-CSF (granulocyte colony-stimulating factor), M-CSF, IL-6 and -7 and LIF (see table 10.1), and many of them are also derived from T-cells and macrophages. It is not surprising therefore that, during a period of chronic inflammation, the cytokines that are produced recruit new precursors into the hematopoietic differentiation pathway—a useful exercise in the circumstances. One of the cytokines, IL-3, should be highlighted for its exceptional ability to support the early cells in this pathway, particularly in synergy with IL-6 and G-CSF.

Network interactions

The complex and integrated relationships between the different cytokines are mediated through cellular events. The genes for IL-3, -4 and -5 and GM-CSF are all tightly linked on chromosome 5 in a region containing genes for M-CSF and its receptor and several other growth factors and receptors. Interaction may occur through a cascade in which one cytokine induces the production of another, through transmodulation of the receptor for another cytokine and through synergism or antagonism of two cytokines acting on the same cell (figure 10.5). The means by which target cells integrate and interpret the complex patterns of stimuli induced by these multiple soluble factors is only slowly unfolding.

DIFFERENT T-CELL SUBSETS CAN MAKE DIFFERENT CYTOKINE PATTERNS

The bipolar Th1/Th2 concept

Helper T-cell clones can be divided into two main types with distinct cytokine secretion phenotypes

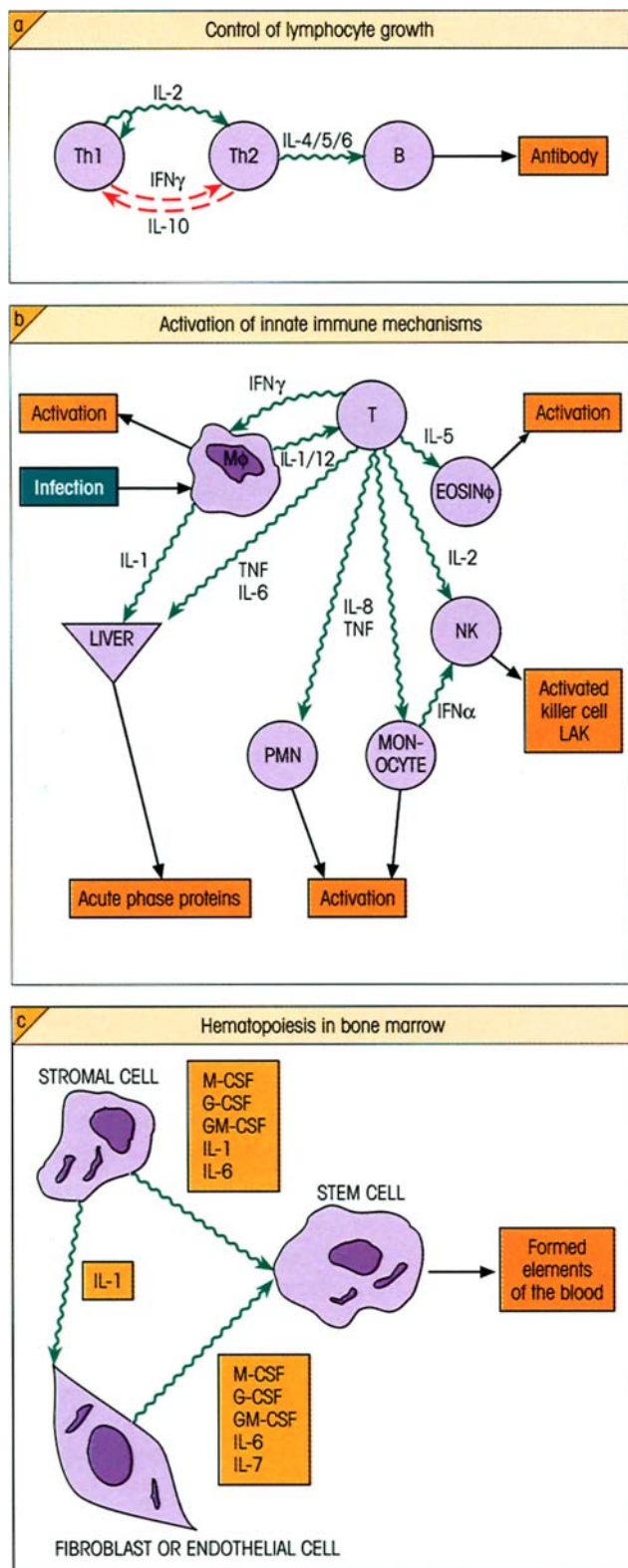


Figure 10.4. Cytokine action. A general but not entirely comprehensive guide to indicate the scope of cytokine interactions (e.g. for reasons of simplicity we have omitted the inhibitory effects of IL-10 on monocytes and the activation of NK cells by IL-12). EOSIN ϕ , eosinophil; LAK, lymphokine-activated killer; M ϕ , macrophage; NK, natural killer cell; PMN, polymorphonuclear neutrophil.

Table 10.2. Cytokine patterns of helper T-cell clones. Interleukin-10 is not listed in the table. Although classed as a Th2 cytokine in the mouse, it is produced by both Th1 and Th2 cells in the human.

CYTOKINE PATTERNS OF T-CELL CLONES		
	Th1	Th2
IFN γ	++	
IL-2	++	
Lymphotoxin (TNF β)	++	
TNF (TNF α)	++	+
GM-CSF	++	+
IL-3	++	+
IL-4		++
IL-5		++
IL-6		++
IL-13		++

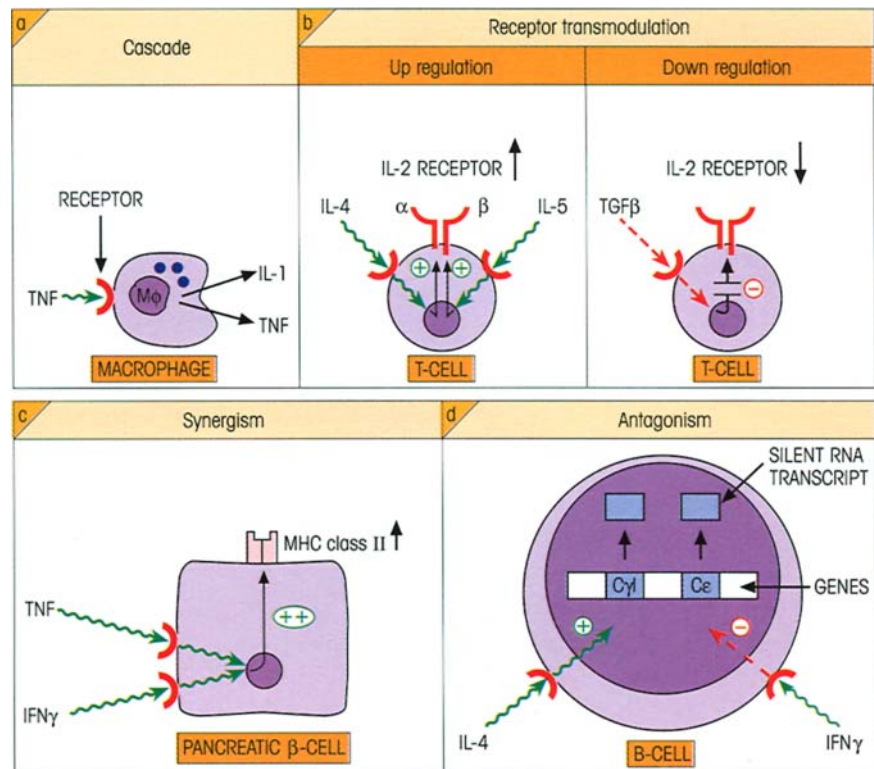
++ ++ + Negative

Th1/2, T-helper 1/2.

(table 10.2). This makes biological sense in that Th1 cells producing cytokines such as IFN γ would be especially effective against intracellular infections with viruses and organisms which grow in macrophages, whereas Th2 cells are very good helpers for B-cells and would seem to be adapted for defense against parasites which are vulnerable to IL-4-switched IgE, IL-5-induced eosinophilia and IL-3/4-stimulated mast cell proliferation. Thus, studies on the infection of mice with the pathogenic protozoan *Leishmania major* demonstrated that intravenous or intraperitoneal injection of killed promastigotes leads to protection against challenge with live parasites associated with high expression of IFN γ mRNA and low levels of IL-4 mRNA; the reciprocal finding of low IFN γ and high IL-4 expression was made after subcutaneous immunization which failed to provide protection. Furthermore, nonvaccinated mice infected with live organisms could be saved by injection of IFN γ and anti-IL-4. These results are consistent with the preferential expansion of a population of protective IFN γ -secreting Th1 cells by intraperitoneal or intravenous immunization, and of nonprotective Th2 cells producing IL-4 in the subcutaneously injected animals. The ability of IFN γ , the characteristic Th1 cytokine, to inhibit the proliferation of Th2 clones, and of Th2-derived IL-4 and -10 to block both proliferation and cytokine release by Th1 cells, would seem to put the issue beyond reasonable doubt (figure 10.6).

The original Mosmann–Coffman classification into Th1 and Th2 subsets was predicated on data obtained

Figure 10.5. Network interactions of cytokines. (a) Cascade: in this example TNF induces secretion of IL-1 and of itself (autocrine) in the macrophage. (Note that all diagrams in this figure are simplified in that the effects on the nucleus are due to messengers resulting from the combination of cytokine with its surface receptor.) (b) Receptor transmodulation showing upregulation of each chain forming the high affinity IL-2 receptor in an activated T-cell by individual cytokines and downregulation by TGF β . (c) Synergy of TNF and IFN γ in upregulation of surface MHC class II molecules on cultured pancreatic insulin-secreting cells. (d) Antagonism of IL-4 and IFN γ on transcription of silent ('sterile') mRNA relating to isotype switch (cf. figure 10.17).



with clones which had been maintained in culture for long periods and might have been artifacts of conditions *in vitro*. The use of cytokine-specific monoclonal antibodies for intracellular fluorescent staining, and of ELISPOT assays (cf. p. 140) for the detection of the secreted molecules, has demonstrated that the Th1/Th2 dichotomy is also apparent in freshly sampled cells and thus also applies *in vivo*. Nonetheless, it is perhaps best not to be too rigidly constrained in one's thinking by the Th1/Th2 paradigm, but rather to look upon T-cells as potentially producing a whole spectrum of cytokine profiles (Th0, figure 10.6), with possible skewing of the responses towards the extreme Th1 and Th2 patterns depending on the nature of the antigen stimulus. Thus, other subsets may also exist, in particular the transforming growth factor- β (TGF β) and IL-10-producing Th3/Tr1 (T-regulatory 1) cells, which are of interest because these cytokines can mediate immunosuppressive effects and may be involved in the induction of mucosally induced tolerance (cf. p. 444).

Interactions with cells of the innate immune system biases the Th1/Th2 response

Antigen-presenting cells, and in particular dendritic cells, appear to be pivotal in driving differentiation towards a Th1 or Th2 phenotype. IL-12 seems to be par-

ticularly important for the production of Th1 cells and IL-4 for the production of Th2 cells. Invasion of phagocytic cells by intracellular pathogens induces copious secretion of IL-12, which in turn stimulates IFN γ production by NK cells. These two cytokines selectively drive differentiation of Th1 development and inhibit Th2 responses (figure 10.6). However, IL-4 effects appear to be dominant over IL-12 and therefore the amounts of IL-4 relative to the amounts of IL-12 and IFN γ will be of paramount importance in determining the differentiation of Th0 cells into Th1 or Th2 (figure 10.7). IL-4 downregulates the expression of the IL-12R β_2 subunit necessary for responsiveness to IL-12, further polarizing the Th2 dominance.

A special cell population, the NK-T cells bearing the NK1.1⁺ marker, rapidly releases an IL-4-dominated pattern of cytokines on stimulation. These cells have many unusual features. They may be CD4⁺8⁻ or CD4⁺8⁺ and express low levels of T-cell $\alpha\beta$ receptors with an invariant α chain and very restricted β , many of these receptors recognizing the nonclassical MHC-like CD1 molecule. Their morphology and granule content are intermediate between T-cells and NK cells. Although they express TCR $\alpha\beta$, there is an inclination to classify them on the fringe of the 'innate' immune system with regard to their primitive characteristics and possession of the lectin-like NK1.1 receptor which

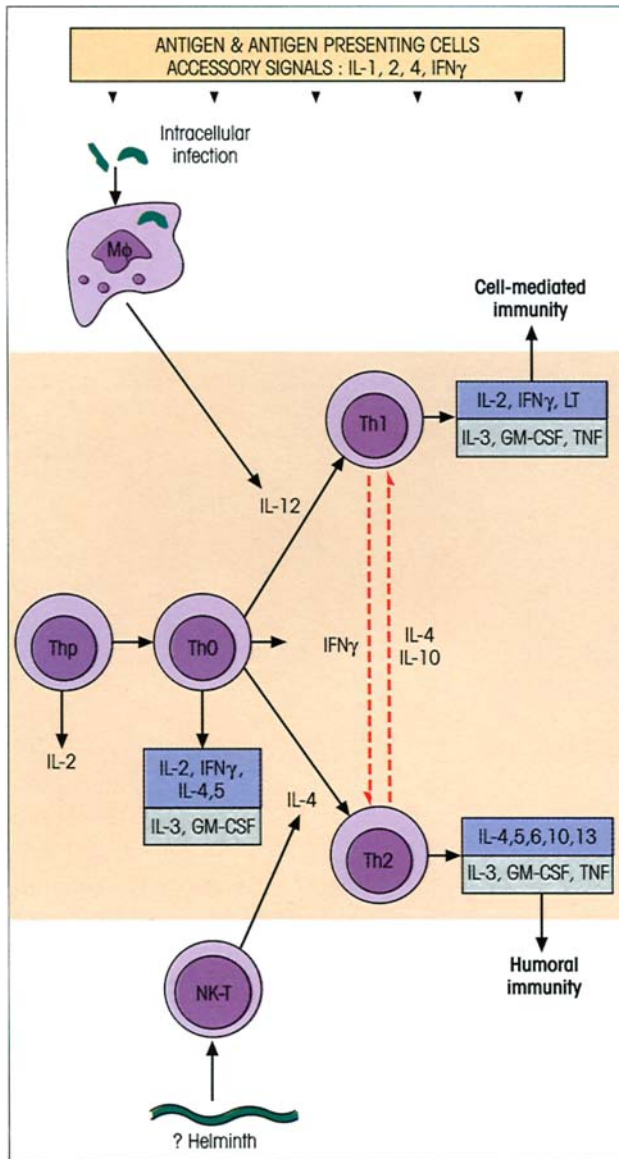


Figure 10.6. The generation of Th1 and Th2 CD4 subsets. It is envisaged that, following the initial stimulation of T-cells, a range of cells producing a spectrum of cytokine patterns emerges. Under different conditions, the resulting population can be biased towards two extremes. IL-12, possibly produced through an 'innate'-type effect of an intracellular infection on macrophages, encourages the development of Th1 cells which produce the cytokines characteristic of *cell-mediated immunity*. IL-4, possibly produced by interaction of microorganisms with the lectin-like NK1.1⁺ receptor on NK-T cells, skews the development to the production of Th2 cells whose cytokines assist the progression of B-cells to antibody secretion and the provision of *humoral immunity*. Cytokines produced by polarized Th1 and Th2 subpopulations are mutually inhibitory. LT, lymphotoxin (TNF β); Thp, T-helper precursor; Th0, early helper cell producing a spectrum of cytokines; other abbreviations as in table 10.1.

may be involved in the recognition of microbial carbohydrates.

Whilst there is a certain amount of evidence indicating the existence of subpopulations of dendritic cells specialized for the stimulation of either Th1 or Th2

populations, this is an area that is still under intensive investigation. However, it should be obvious from the above discussion that the cytokines produced in the immediate vicinity of the T-cell will be important. To give one recent example, Cantor and colleagues homologously deleted the gene for the cytokine Eta-1 (osteopontin) in mice and found that these animals had severely impaired immunity to infection with herpes simplex virus and to the intracellular bacterium *Listeria monocytogenes*. This was due to a deficient Th1 immunity caused by reduced IL-12 and IFN γ and enhanced IL-10 production. It appears that Eta-1 production by activated T-cells stimulates IL-12 production by macrophage lineage cells and downregulates IL-10 production. Interestingly, both serine phosphorylated and nonphosphorylated forms of Eta-1 are secreted by T-cells, and the IL-12 effect is phosphorylation-dependent whereas the IL-10 effect is not, indicating that phosphorylation can regulate the activity of secreted proteins.

Cytotoxic T-cells can also be subdivided into Tc1/Tc2

Clones of human cytotoxic T-cells obtained by limiting dilution also characteristically secrete particular cytokines. Thus, Tc1 cells secrete IFN γ but not IL-4, whilst Tc2 cells secrete IL-4 but not IFN γ . These clones show no differences in their cytolytic function but, when cocultured with CD4⁺ T-cells, Tc1 clones induce Th1 cells, whilst Tc2 clones induce Th2 cells.

ACTIVATED T-CELLS PROLIFERATE IN RESPONSE TO CYTOKINES

In so far as T-cells are concerned, amplification following activation is critically dependent upon IL-2 (figure 10.8). This cytokine is a single peptide of molecular weight 15.5 kDa which acts only on cells which express high affinity IL-2 receptors (figure 10.2). These receptors are not present on resting cells, but are synthesized within a few hours after activation.

Separation of an activated T-cell population into those with high and low affinity IL-2 receptors showed clearly that an adequate number of high affinity receptors were mandatory for the mitogenic action of IL-2. The numbers of these receptors on the cell increase under the action of antigen and of IL-2 and, as antigen is cleared, so the receptor numbers decline and, with that, the responsiveness to IL-2. It should be appreciated that, although IL-2 is an immunologically nonspecific T-cell growth factor, it only functions appropriately in specific responses because unstimulated T-cells do not express IL-2 receptors.

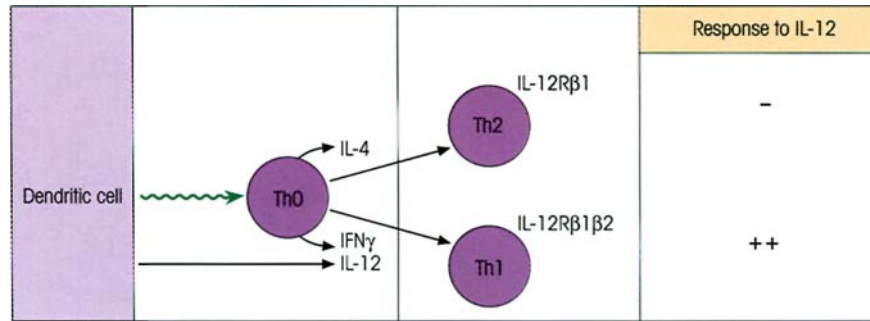


Figure 10.7. Polarization of T-helper responses is driven by dendritic cells. Initial activation of the Th0 cell is dependent upon presentation of peptide–MHC to the TCR, together with costimulatory signals such as that provided by CD80 and CD86 (B7.1 and B7.2 respectively) on the dendritic cell. Subsequently, the T-cells are driven towards a Th2 phenotype by the presence of IL-4 or a Th1 phenotype

by the presence of IL-12 and IFN γ . The Th0 cells express both the β_1 and β_2 chains of the IL-12 receptor, but IL-4 causes loss of expression of the β_2 chain. Whilst IL-12 is predominantly a product of the macrophage/dendritic cell lineage, IL-4 is produced by a number of cell types including mast cells, $\gamma\delta$ T-cells, NK-T cells, Th0 and Th2 cells.

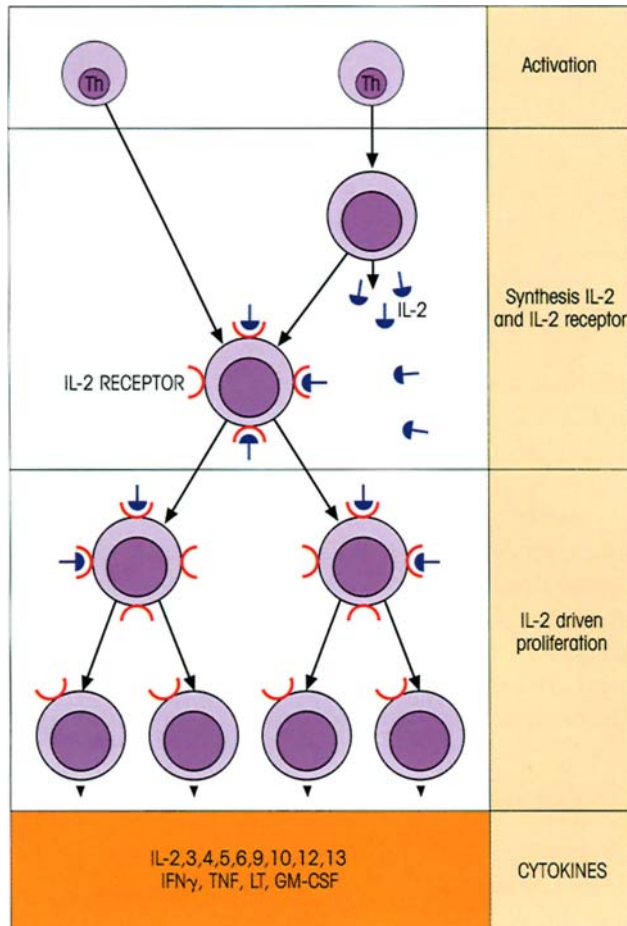


Figure 10.8. Activated T-blasts expressing surface receptors for IL-2 proliferate in response to IL-2 produced by itself or by another T-cell subset. Expansion is controlled through downregulation of the IL-2 receptor by IL-2 itself. The expanded population secretes a wide variety of biologically active cytokines of which IL-4 also enhances T-cell proliferation.

The T-cell blasts also produce an impressive array of other cytokines, and the proliferative effect of IL-2 is reinforced by the action of IL-4 and, to some extent, IL-6, which react with corresponding receptors on the dividing T-cells. We must not lose sight of the importance of control mechanisms, and obvious candidates to subsume this role are TGF β , which blocks IL-2-induced proliferation (figure 10.5b) and the production of TNF (TNF α) and lymphotoxin (TNF β), and the cytokines IFN γ , IL-4 and IL-12, which mediate the mutual antagonism of Th1 and Th2 subsets.

T-CELL EFFECTORS IN CELL-MEDIATED IMMUNITY

Cytokines mediate chronic inflammatory responses

In addition to their role in the adaptive response, T-cell cytokines are responsible for generating antigen-specific chronic inflammatory reactions which deal with intracellular parasites (figure 10.4b and 10.9), although there is a different emphasis on the pattern of factors involved (cf. p. 274).

Early events

The initiating event is probably a local inflammatory response to tissue injury caused by the infectious agent which would upregulate the synthesis of adhesion molecules such as VCAM-1 (vascular cell adhesion molecule) and ICAM-1 on adjacent vascular endothelial cells. These would permit entry of memory T-cells to the infected site through their VLA-4 and LFA-1 homing receptors (cf. p. 152). Contact with processed

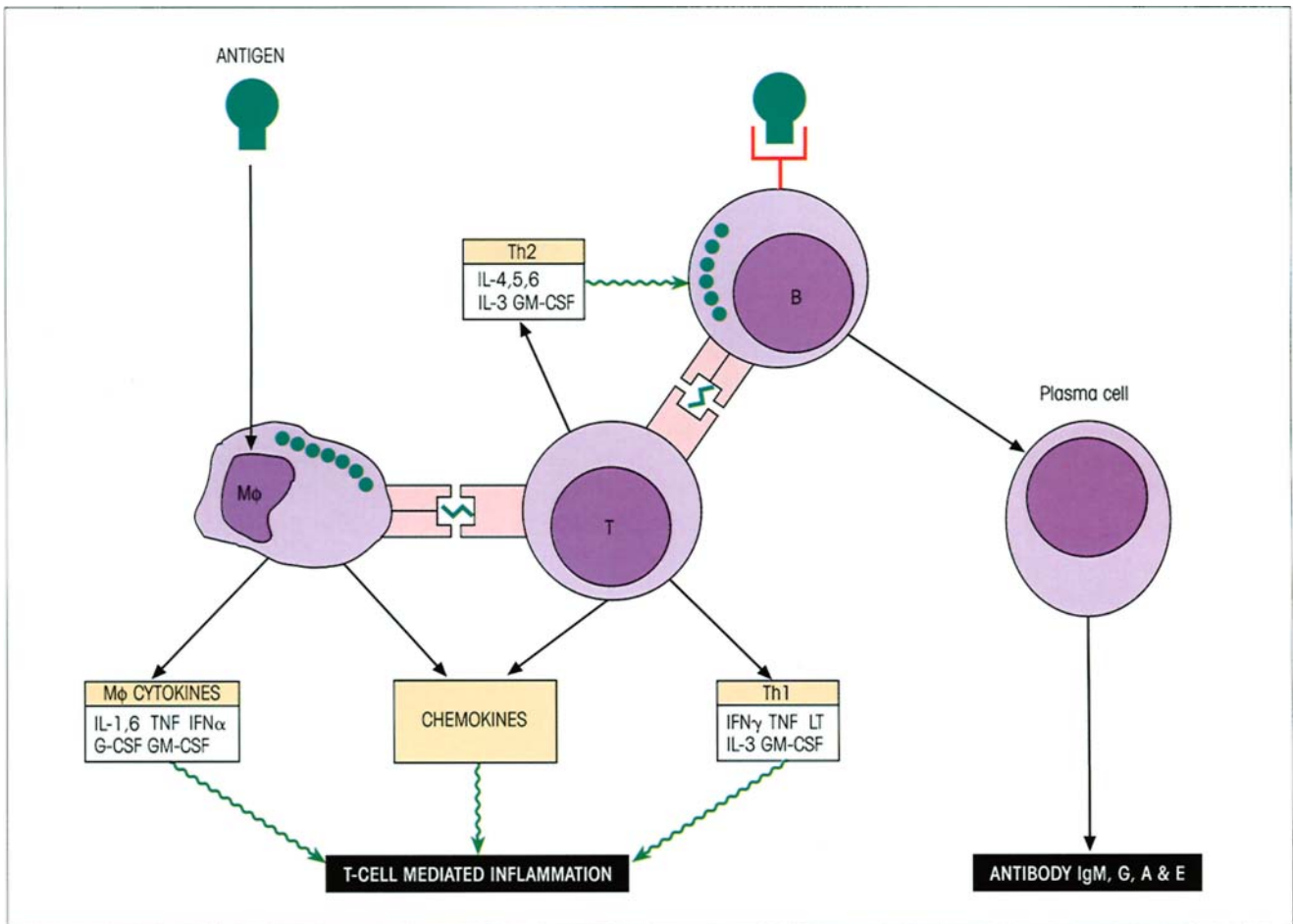


Figure 10.9. Cytokines controlling the antibody and T-cell-mediated inflammatory responses. Abbreviations as in table 10.1.

antigen derived from the intracellular parasite will activate the specific T-cell and induce the release of secreted cytokines. TNF will further enhance the expression of endothelial accessory molecules and increase the chances of other memory cells in the circulation homing in to meet the antigen provoking inflammation.

Chemotaxis

The recruitment of T-cells and macrophages to the inflammatory site (figure 10.9) is greatly enhanced by the action of chemotactic cytokines termed **chemokines** (*chemoattractant cytokine*). These can be potentially produced by a variety of cell types and are divided into four families based on the disposition of the first (N-terminal) two of the four canonical cysteine residues (table 10.3). CXC chemokines have one amino acid and CX3C have three amino acids between the two cysteines. CC chemokines have adjacent cysteines at this

location, whereas C chemokines lack cysteines 1 and 3 found in other chemokines. Chemokines bind to G-protein-coupled seven transmembrane receptors (figure 10.2). Despite the fact that a single chemokine can sometimes bind to more than one receptor, and a single receptor can bind several chemokines, many chemokines exhibit a strong tissue and receptor specificity. They play important roles in inflammation, lymphoid organ development, cell trafficking, cell compartmentalization within lymphoid tissues, Th1/Th2 development, angiogenesis and wound healing.

Macrophage activation

Macrophages with intracellular organisms are activated by agents such as IFN γ , GM-CSF, IL-2 and TNF and should become endowed with microbicidal powers. During this process, some macrophages may die (probably helped along by cytotoxic T-cells) and release living parasites, but these will be dealt with by fresh macrophages brought to the site by chemotaxis and newly activated by local cytokines so that they have passed the stage of differentiation at which the

Table 10.3. Chemokines and their receptors. The chemokines are grouped according to the arrangement of their cysteines (see text). The letter L designates ligand (i.e. the individual chemokine), whereas the letter R designates receptors. Names in parentheses refer to the murine homologs of the human chemokine where the names of these differ, or the murine chemokine alone if no human equivalent has been described.

FAMILY	CHEMOKINE	ALTERNATIVE NAMES	CHEMOTAXIS	RECEPTORS
CXC	CXCL1	GRO α /MGS α	Neutro	CXCR2 > CXCR1
	CXCL2	GRO β /MGS β	Neutro	CXCR2
	CXCL3	GRO γ /MGS γ	Neutro	CXCR2
	CXCL4	PF4	Eosino, Baso	?
	CXCL5	ENA-78	Neutro	CXCR2
	CXCL6	GCP-2/(CK α -3)	Neutro	CXCR1, CXCR2
	CXCL7	NAP-2	Neutro	CXCR2
	CXCL8	IL-8	Neutro	CXCR1, CXCR2
	CXCL9	Mig	T, NK	CXCR3
	CXCL10	IP-10	T, NK	CXCR3
	CXCL11	I-TAC	T, NK	CXCR3
	CXCL12	SDF-1 α / β	T, B, DC, Mono	CXCR4
	CXCL13	BLC/BCA-1	B	CXCR5
	CXCL14	BRAC/Bolekin	?	?
	CXCL15	Lungkine	Neutro	?
C	XCL1	Lymphotactin/SCM-1 α /ATAC	T	XCR1
	XCL2	SCM-1 β	T	XCR1
CX3C	CX3CL1	Fractalkine/Neurotactin	T, NK, Mono	CX3CR1
CC	CCL1	I-309/(TCA-3/P500)	Mono	CCR8
	CCL2	MCP-1/MCAF	T, NK, DC, Mono, Baso	CCR2
	CCL3	MIP-1 α /LD78 α	T, NK, DC, Mono, Eosino	CCR1, CCR5
	CCL4	MIP-1 β	T, NK, DC, Mono	CCR5
	CCL5	RANTES	T, NK, DC, Mono, Eosino, Baso	CCR1, CCR3, CCR5
	(CCL6)	(C10/MRP-1)	?	?
	CCL7	MCP-3	T, NK, DC, Mono, Eosino, Baso	CCR1, CCR2, CCR3
	CCL8	MCP-2	T, NK, DC, Mono, Baso	CCR3
	(CCL9/10)	(MRP-2/CCF18/MIP-1 γ)	?	?
	CCL11	Eotaxin-1	T, DC, Eosino, Baso	CCR3
	(CCL12)	(MCP-5)	T, NK, DC, Mono, Baso	CCR2
	CCL13	MCP-4	T, NK, DC, Mono, Eosino, Baso	CCR2, CCR3
	CCL14	HCC-1/HCC-3	T, Mono, Eosino	CCR1
	CCL15	HCC-2/Leukotactin-1/MIP-1 δ	T	CCR1, CCR3
	CCL16	HCC-4/LEC/(LCC-1)	T	CCR1
	CCL17	TARC	T, DC, Mono	CCR4
	CCL18	DCCK1/PARC/AMAC-1	T	?
	CCL19	MIP-3 β /ELC/Exodus-3	T, B, DC	CCR7
	CCL20	MIP-3 α /LARC/Exodus-1	DC	CCR6
	CCL21	6Ckine/SLC/Exodus-2/(TCA-4)	T, DC	CCR7
	CCL22	MDC/STCP-1/ABCD-1	T, DC, Mono	CCR4
	CCL23	MPIF-1	T	CCR1
	CCL24	MPIF-2/Eotaxin-2	T, DC, Eosino, Baso	CCR3
	CCL25	TECK	T, DC, Mono	CCR9
	CCL26	SCYA26/Eotaxin-3	T	CCR3
	CCL27	CTACK/ALP/ESkine	T	CCR10

B, B-cell; Baso, basophil; DC, dendritic cell; Eosino, eosinophil; Mono, monocyte; Neutro, neutrophil; NK, natural killer; T, T-cell.

intracellular parasites can subvert their killing mechanisms (cf. p. 263).

Combating viral infection

Virally infected cells require a different strategy and one strand of that strategy exploits the innate interferon mechanism to deny the virus access to the cell's replicative machinery. IFN γ , TNF and lymphotoxin all

induce 2'-5' (A) synthetase, a protein which is involved in viral protection. TNF has another string to its bow in its ability to kill certain cells, since death of an infected cell before viral replication has occurred is obviously beneficial to the host. Its cytotoxic potential was first recognized using tumor cells as targets (hence the name), and IFN γ and lymphotoxin can act synergistically, with IFN γ setting up the cell for destruction by inducing the formation of TNF receptors.

Killer T-cells

The generation of cytotoxic T-cells

Cytotoxic T-cells (Tc), also referred to as cytotoxic T-lymphocytes (CTLs), represent the other major arm of the cell-mediated immune response and are of strategic importance in the killing of virally infected cells and possibly in contributing to the postulated surveillance mechanisms against cancer cells.

The cytotoxic cell precursors recognize antigen on the surface of cells in association with class I major histocompatibility complex (MHC) molecules and, like B-cells, they usually require help from T-cells. The mechanism by which help is proffered may, however, be quite different. As explained earlier (see p. 173), effective T-B collaboration is usually 'cognate' in that the collaborating cells recognize two epitopes which are physically linked (usually on the same molecule). If we may remind the reader without causing offense, the reason for this is that the surface Ig receptors on the B-cell capture native antigen, process it internally and present it to the Th as a peptide in association with MHC class II. Although it has been shown that linked epitopes on the antigen are also necessary for cooperation between Th and the cytotoxic T-cell precursor (Tc), the nature of T-cell recognition prevents native antigen being focused onto the Tc by its receptor for subsequent processing, even if that cell were to express MHC II, which in its resting state it does not. It seems most likely that Th and Tc bind to the same APC, for example a dendritic cell, which has processed viral antigen and displays processed viral peptides in association with both class II (for the Th cell) and class I (for the Tc) on its surface; one cannot exclude the possibility that the APC could be the virally infected cell itself. Cytokines from the triggered Th will be released in close proximity to the Tc which is engaging the antigen-MHC signal and will be stimulated to proliferate and differentiate into a Tc under the influence of IL-2 and -6 (figure 10.10a). However, interaction of the APC with the Th and the Tc cell can be temporally separated and, in this case, it appears that the helper T-cell 'licenses' the dendritic cell for future interaction with the cytotoxic T-cell. It does this by activating the dendritic cell through CD40, thereby upregulating costimulatory molecules and cytokine production, in particular IL-12, by the dendritic cell (figure 10.10b). An entirely Th-independent mechanism of Tc activation is also thought to occur. This has been demonstrated in, for example, the response to protein antigens given with potent adjuvants such as immunostimulatory DNA sequences (ISSs), in this case

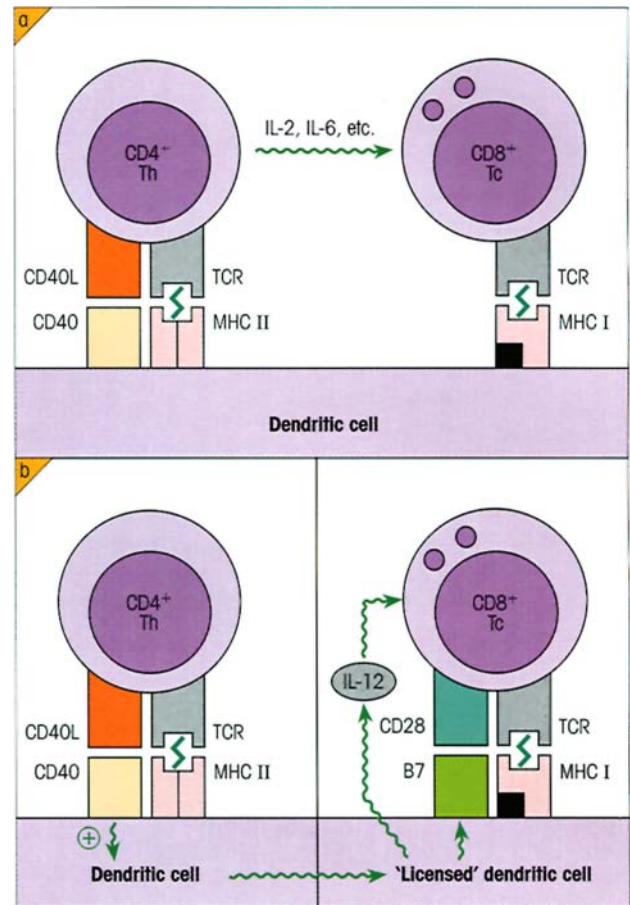


Figure 10.10. T-helper cell activation of cytotoxic T-cells. Activation of the CD4⁺ helper T-cells (Th) by the dendritic cell involves a CD40-CD40 ligand (CD154) costimulatory signal and recognition of an MHC class II peptide presented by the T-cell receptor. (a) If both the Th and the cytotoxic T-lymphocyte (Tc) are present at the same time, the release of cytokines from the activated Th cells stimulates the differentiation of the CD8⁺ precursor into an activated, MHC class I-restricted Tc. However, as shown in (b), the Th and the Tc do not need to interact with the APC at the same time. In this case, the Th cell 'licenses' the dendritic cell for future interaction with a Tc cell. Thus the Th cell, by engaging CD40, drives the dendritic cell from a resting state into an activated state with upregulation of costimulatory molecules such as B7.1 and B7.2 (CD80 and CD86, respectively) and increased cytokine production, particularly of IL-12.

possibly involving adjuvant-induced production of proinflammatory cytokines and cell surface costimulatory molecules.

The lethal process

Cytotoxic T-cells (Tc) are generally of the CD8 subset, and their binding to the target cell through T-cell receptor recognition of antigen plus class I MHC is assisted by association between CD8 and class I and by other accessory molecules such as LFA-1 and CD2 (see figure 9.3).

Tc are **unusual secretory cells** which use a modified lysosome to secrete their lytic proteins. Following delivery of the TCR/CD3 signal, the **lytic granules** are driven at a rare old speed (up to $1.2\mu\text{m}/\text{sec}$) along the microtubule system and delivered to the point of contact between the Tc and its target (figure 10.11). This guarantees the specificity of killing dictated by TCR recognition of the target and limits any collateral damage to surrounding cells. As with NK cells, which have comparable granules (cf. p. 18), exocytosis of the granule contents, including perforins, granzymes and TNF, causes lesions in the target cell membrane and death by inducing apoptosis. Tc are endowed with a second killing mechanism involving Fas and its ligand (cf. p. 19), but the inability of perforin knockout mice to clear viruses effectively suggests that the secretory

granules provide the dominant attack on virally infected cells. Videomicroscopy shows that Tc are serial killers. After the 'kiss of death', the T-cell can disengage and seek a further victim, there being rapid synthesis of new granules.

One should also not lose sight of the fact that CD8 cells synthesize other cytokines such as $\text{IFN}\gamma$ which also have antiviral potential.

Inflammation must be regulated

Once the inflammatory process has cleared the inciting agent, the body needs to switch it off. IL-10 has profound anti-inflammatory and immunoregulatory effects, acting on macrophages and Th1 cells to inhibit release of factors such as IL-1 and TNF. It induces the release of soluble TNF receptors which are endogenous inhibitors of TNF, and downregulates surface TNF receptor. Soluble IL-1 receptors released during inflammation can act to 'decoy' IL-1 itself. IL-4 not only acts to constrain Th1 cells but also upregulates production of the natural inhibitor of IL-1, the IL-1 receptor antagonist (IL-1Ra). The role of $\text{TGF}\beta$ is more difficult to tease out because it has some pro- and other anti-inflammatory effects, although it undoubtedly promotes tissue repair after resolution of the inflammation.

PROLIFERATION AND MATURATION OF B-CELL RESPONSES ARE MEDIATED BY CYTOKINES

The activation of B-cells by Th, through the TCR recognition of MHC-linked antigenic peptide plus the costimulatory **CD40L–CD40 interaction**, leads to upregulation of the surface receptor for IL-4. Copious local release of this cytokine from the Th then drives powerful clonal proliferation and expansion of the activated B-cell population. IL-2 and IL-13 also contribute to this process (figure 10.12).

Under the influence of IL-4 and IL-13, the expanded clones can differentiate and mature into IgE synthesizing cells. $\text{TGF}\beta$ and IL-5 encourage cells to switch their Ig class to IgA. IgM plasma cells emerge under the tutelage of IL-4 plus -5, and IgG producers result from the combined influence of IL-4, -5, -6, -13 and $\text{IFN}\gamma$ (figure 10.12).

Type 2 thymus-independent antigens can activate B-cells directly (cf. p. 171) but nonetheless still need cytokines for efficient proliferation and Ig production. These may come from accessory cells such as NK and NK-T cells which bear lectin-like receptors.

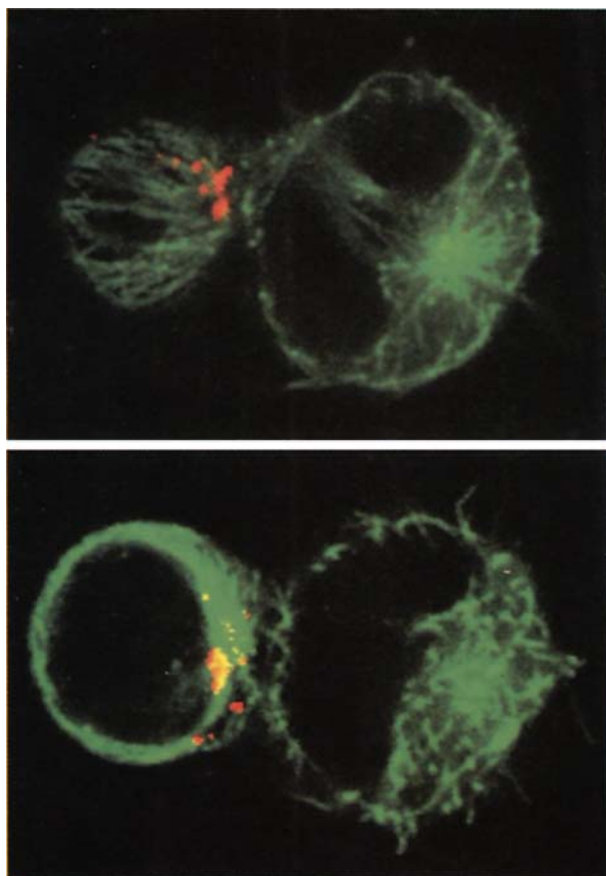


Figure 10.11. Conjugation of a cytotoxic T-cell (on left) to its target, here a mouse mastocytoma, showing polarization of the granules towards the target at the point of contact. The cytoskeletons of both cells are revealed by immunofluorescent staining with an antibody to tubulin (green) and the lytic granules with an antibody to granzyme A (red). Twenty minutes after conjugation the target cell cytoskeleton may still be intact (above), but this rapidly becomes disrupted (below). (Photographs kindly provided by Dr Gillian Griffiths.)

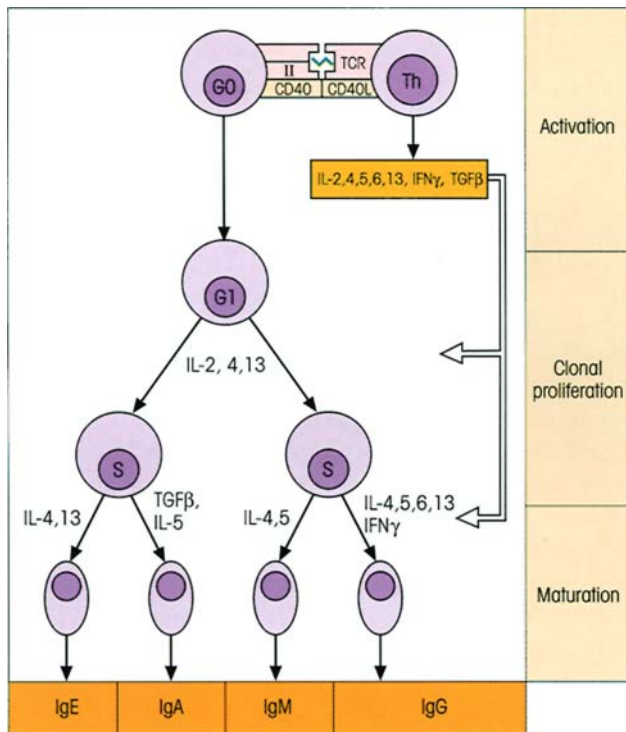


Figure 10.12. B-cell response to thymus-dependent (TD) antigen: clonal expansion and maturation of activated B-cells under the influence of T-cell-derived soluble factors. Costimulation through the CD40L–CD40 interaction is essential for primary and secondary immune responses to TD antigens and for the formation of germinal centers and memory. *c-myc* expression, which is maximal 2 hours after antigen or anti- μ stimulation, parallels sensitivity to growth factors; transfection with *c-myc* substitutes for anti- μ .

WHAT IS GOING ON IN THE GERMINAL CENTER?

The secondary follicle with its corona or mantle of small lymphocytes surrounding the pale germinal center is a striking and unique cellular structure. First, let us recall the overall events described in Chapter 8. Secondary challenge with antigen or immune complexes induces enlargement of germinal centers, formation of new ones, appearance of memory B-cells and development of Ig-producing cells of higher affinity. B-cells entering the germinal center become **centroblasts** which divide with a very short cycle time of 6 hours, and then become nondividing **centrocytes** in the basal light zone, many of which die from apoptosis (figure 10.13). As the surviving centrocytes mature, they differentiate either into **immunoblast plasma cell precursors**, which secrete Ig in the absence of antigen, or into **memory B-cells**.

What then is the underlying scenario? Following secondary antigen challenge, primed B-cells may be activated by paracortical Th cells in association with interdigitating dendritic cells or macrophages, and

migrate to the germinal center. There they divide in response to powerful stimuli from complexes on follicular dendritic cells (cf. p. 160) and from cytokines released by T-cells in response to antigen-presenting B-cells. During this particularly frenetic bout of cell division, **somatic hypermutation** of B-cell Ig genes occurs. The cells also undergo **Ig class switching**. Thereafter, as they transform to centrocytes, they are vulnerable and die readily, whence they are taken up as the 'tingible bodies' by macrophages, unless rescued by association with antigen on a follicular dendritic cell. This could result from cross-linking of surface Ig receptors and is accompanied by expression of *bcl-2* which protects against apoptosis. Signaling through CD40, by presentation of antigen to Th cells, would also prolong the life of the centrocyte. In either case, the interactions will only occur if the mutated surface Ig receptor still binds antigen and, as the concentration of antigen gradually falls, only if the receptor is of high affinity. In other words, the system can deliver high affinity antibody by a Darwinian process of high frequency mutation of the Ig genes and selection by antigen of the cells bearing the antibody which binds most strongly (figure 10.14). This increase of affinity as the antibody level falls late in the response is of obvious benefit, since a small amount of high affinity antibody can do the job of a large amount of low affinity (as in boxing, a small 'goodun' will generally be a match for a mediocre 'bigun').

Further differentiation now occurs. The cells either migrate to the sites of plasma cell activity (e.g. lymph node medulla) or go to expand the memory B-cell pool depending upon the cytokine and other signals received. CD40 engagement by CD40 ligand on a T-cell guides the B-cell into the memory compartment.

THE SYNTHESIS OF ANTIBODY

The sequential processes by which secreted Ig arises are illustrated in figure 10.15. In the normal antibody-forming cell there is a rapid turnover of light chains which are present in slight excess. Defective control occurs in many myeloma cells and one may see excessive production of light chains or complete suppression of heavy chain synthesis.

The variable and constant regions are spliced together in the mRNA before leaving the nucleus. Differential splicing mechanisms also provide a rational explanation for the coexpression of surface IgM and IgD with identical V regions on a single cell, and for the switch from production of membrane-bound IgM receptor to secretory IgM in the antibody-forming cell (cf. figures 4.1 and 4.2).

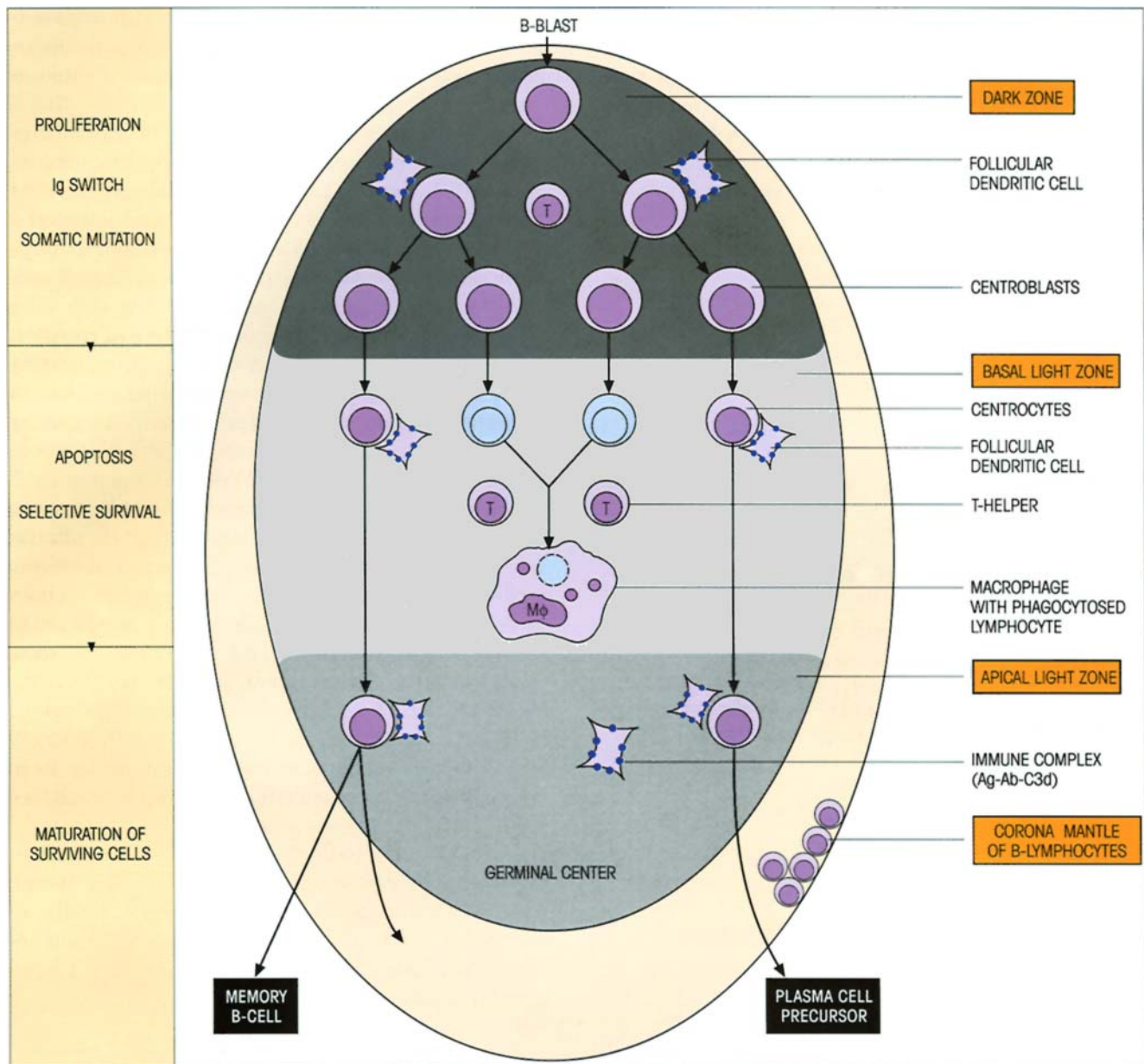


Figure 10.13. The events occurring in lymphoid germinal centers. Germinal center B-cells can be enriched through their affinity for the peanut agglutinin lectin. They show numerous mutations in the antibody genes. Expression of LFA-1 and ICAM-1 on B-cells and follicular dendritic cells (FDCs) in the germinal center makes them 'sticky'. Centroblasts at the base of the follicle are strongly CD77 positive. The Th cells bear the unusual CD57 marker. The FDCs all express CD21 and CD54; those in the apical light zone are strongly CD23 positive, those in the basal light zone express little CD23.

Through their surface receptors, FDCs bind immune complexes containing antigen and C3 which, in turn, are very effective B-cell stimulators since coligation of the surface receptors for antigen and C3 (CR2) lowers their threshold for activation. The costimulatory molecules CD40 and B7 play pivotal roles. Antibodies to CD40 prevent formation of germinal centers and anti-CD40L can disrupt established germinal centers within 12 hours. Anti-B7.2, given early in the immune response, prevents germinal center formation and, when given at the onset of hypermutation, suppresses that process.

IMMUNOGLOBULIN CLASS SWITCHING OCCURS IN INDIVIDUAL B-CELLS

The synthesis of antibodies belonging to the various immunoglobulin classes proceeds at different rates. Usually there is an early IgM response which tends to

fall off rapidly. IgG antibody synthesis builds up to its maximum over a longer time period. On secondary challenge with antigen, the time-course of the IgM response resembles that seen in the primary. By contrast, the synthesis of IgG antibodies rapidly accelerates to a much higher titer and there is a relatively slow fall-off

in serum antibody levels (figure 10.16). The same probably holds for IgA, and in a sense both these immunoglobulin classes provide the main immediate defense against future penetration by foreign antigens.

Individual cells can switch over from IgM to IgG production. For example, antigen challenge of irradiated recipients receiving relatively small numbers of lymphoid cells produced splenic foci of cells, each synthesizing antibodies of different heavy chain class

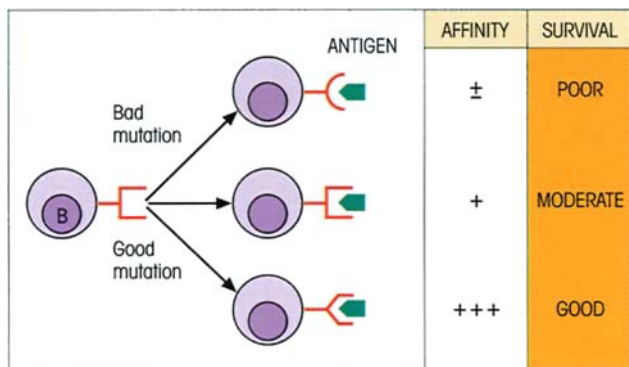


Figure 10.14. Darwinian selection by antigen of B-cells with antibody mutants of high affinity protects against cell death in the germinal center, either through cross-linking of sIg by antigen on follicular dendritic cells, or through Th cell recognition of processed antigen and signaling through CD40. In both cases, capture of antigen, particularly as the concentration falls, will be critically affected by the affinity of the surface receptor.

bearing a single idotype; the common idotype suggests that each focus is derived from a single precursor cell whose progeny can form antibodies of different class.

Antibody synthesis in most classes shows considerable dependence upon T cooperation in that the responses in T-deprived animals are strikingly deficient; such is true of mouse IgG1, IgG2a, IgA, IgE and part of the IgM antibody responses. T-independent antigens such as the polyclonal activator, lipopolysaccharide (LPS) endotoxin, induce synthesis of IgM with some IgG2b and IgG3. Immunopotentiality by complete Freund's adjuvant, a water-in-oil emulsion containing antigen in the aqueous phase and a suspension of killed tubercle bacilli in the oil phase (see p. 300), seems to occur, at least in part through the activation of Th cells which stimulate antibody production in T-dependent classes. The prediction from this, that the response to T-independent antigens (e.g. *Pneumococcus* polysaccharide, p. 171) should not be potentiated by Freund's adjuvant, is borne out in practice; furthermore, as mentioned previously, these antigens evoke primarily IgM antibodies and poor immunological memory, as do T-dependent antigens injected into T-cell-deficient, neonatally thymectomized hosts.

Thus, in rodents at least, the switch from IgM to IgG and other classes appears to be largely under T-cell control critically mediated by CD40 and by cytokines

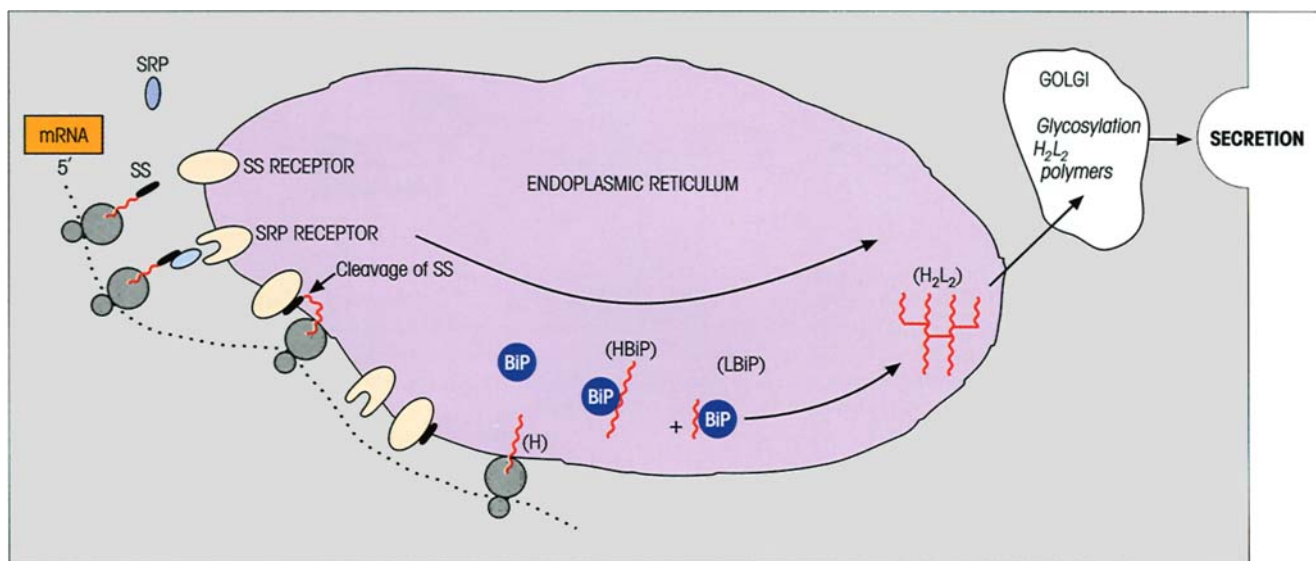


Figure 10.15. Synthesis of immunoglobulin. As mRNA is translated on the ribosome, the N-terminal signal sequence (SS) is bound by a signal recognition particle (SRP) which docks onto a receptor on the outer membrane of the endoplasmic reticulum (ER) and facilitates entry of the nascent Ig chain into the ER lumen. The SS associates with a specific membrane receptor and is cleaved; the remainder of the chain, as it elongates, complexes with the molecular chaperone

BiP (heavy chain-binding protein) which binds to the heavy chain C_{H1} and V_L domains to control protein folding. The unassembled chains oxidize and dissociate as the full H_2L_2 Ig molecule. The assembled H_2L_2 molecules can now leave the ER for terminal glycosylation in the Golgi and final secretion. Surface receptor Ig would be inserted by its hydrophobic sequences into the membrane of the endoplasmic reticulum as it was synthesized.

(see p. 189). Let us take another look at the stimulation of small, surface IgM-positive, B-cells by LPS. As we noted, on its own, the nonspecific mitogen evokes the synthesis of IgM, IgG3 and some IgG2b. Following addition of IL-4 to the system, there is class switching from IgM to IgE and IgG1 production, whereas IFN γ stimulates class switching from IgM to IgG2a and TGF β induces switching from IgM to IgA or IgG2b.

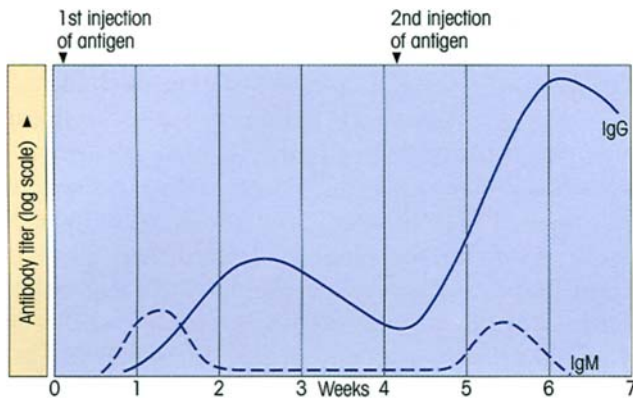


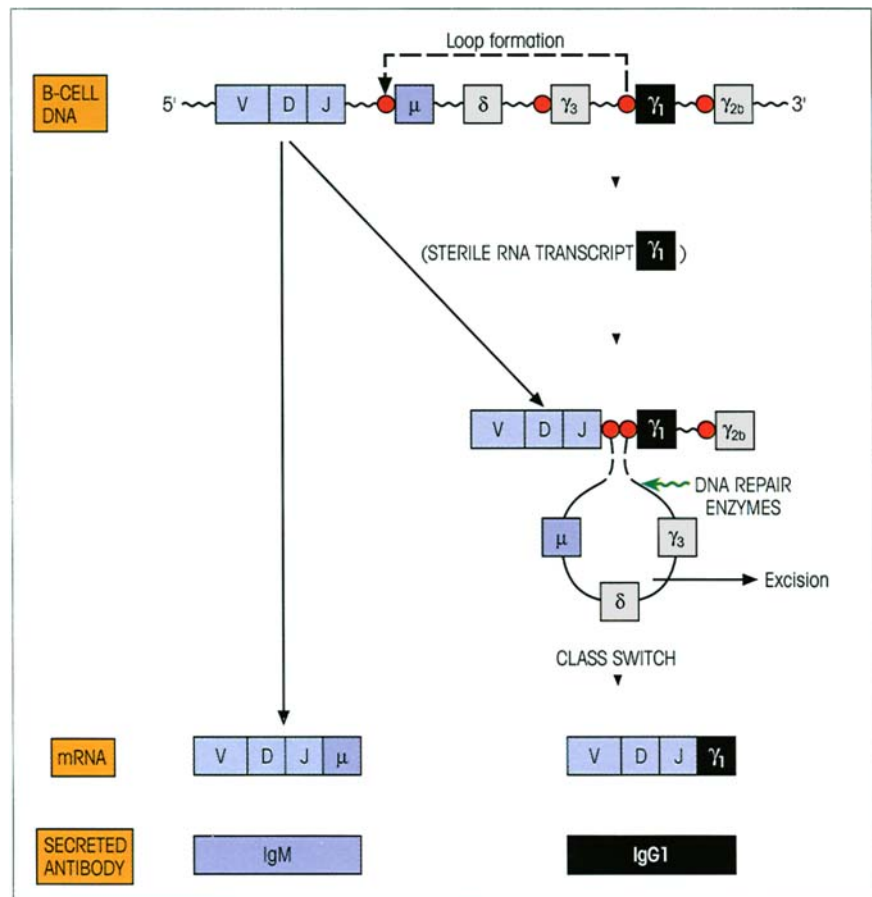
Figure 10.16. Synthesis of IgM and IgG antibody classes in the primary and secondary responses to antigen.

These cytokines induce the formation of germ-line sterile transcripts which start at the I (initiation) exon 5' of the switch region for the antibody class to which switching will occur and terminate at the polyadenylation site 3' of the relevant C_H gene (figure 10.17). The transcripts are not translated but instead remain associated with the template DNA, forming RNA–DNA hybrids within the S regions of the DNA which might act as targets for enzymes involved in the recombination process. Under the influence of the recombinase, a given VDJ gene segment is transferred from $\mu\delta$ to the new constant region gene (figure 10.17), so yielding antibodies of the same specificity but of different class.

Class-switched B-cells are subject to high mutation rates after the initial response

The reader will no doubt recollect that this idea was raised in Chapter 4 when discussing the generation of diversity, and that the germinal center has been identified as the site of intense mutagenesis. The normal V-region mutation rate is of the order of 10^{-5} /base pair/cell division, but this rises to

Figure 10.17. Class switching to produce antibodies of identical specificity but different immunoglobulin isotype (in this example from IgM to IgG1) is achieved by a recombination process which utilizes the specialized switch sequences (●) and leads to a loss of the intervening DNA loop (μ , δ and γ_3). Each switch sequence is 1–10 kilobases in length and comprises guanosine-rich repeats of 20–100 base pairs. Because the switch sequence associated with each C_H gene has a unique nucleotide sequence, recombination cannot occur homologously and therefore probably depends upon nonhomologous end joining. DNA repair proteins including Ku70, Ku80 and the catalytic subunit of the DNA-dependent protein kinase (DNA-PK $_{CS}$) are involved in this process.



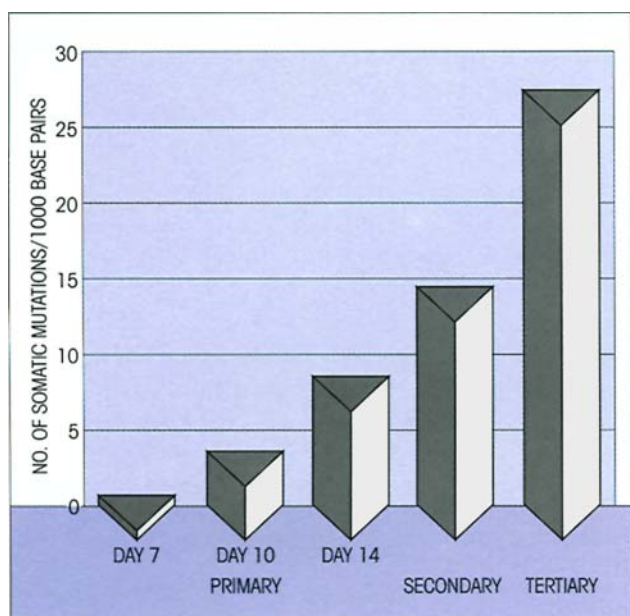


Figure 10.18. Increasing somatic mutations in the immunodominant germ-line antibody observed in hybridomas isolated following repeated immunization with phenyloxazolone. (Data from Berek C. & Apel M. (1989) In Melchers F. *et al.* (eds) *Progress in Immunology* 7, 99. Springer-Verlag, Berlin.)

10^{-3} /base pair/generation in B-cells as a result of antigenic stimulation. This process is illustrated in figure 10.18 which charts the accumulation of somatic mutations in the immunodominant V_H/V_K antibody structure during the immune response to phenyloxazolone. With time and successive boosting, the mutation rate is seen to rise dramatically and, in the context of the present discussion, it is clear that the strategically targeted hypermutations occurring within or adjacent to the complementarity determining hypervariable loops (figure 10.19) can give rise to cells which secrete antibodies having a different combining affinity to that of the original parent cell. Randomly, some mutated daughter cells will have higher affinity for antigen, some the same or lower and others perhaps none at all (cf. figure 10.14). Similarly, mutations in the framework regions may be 'silent' or, if they disrupt the folding of the protein, give rise to nonfunctional molecules. Pertinently, the proportion of germinal center B-cells with 'silent' mutations is high early in the immune response but falls dramatically with time, suggesting that early diversification is followed by preferential expansion of clones expressing mutations which improve their chances of reacting with and being stimulated by antigen.

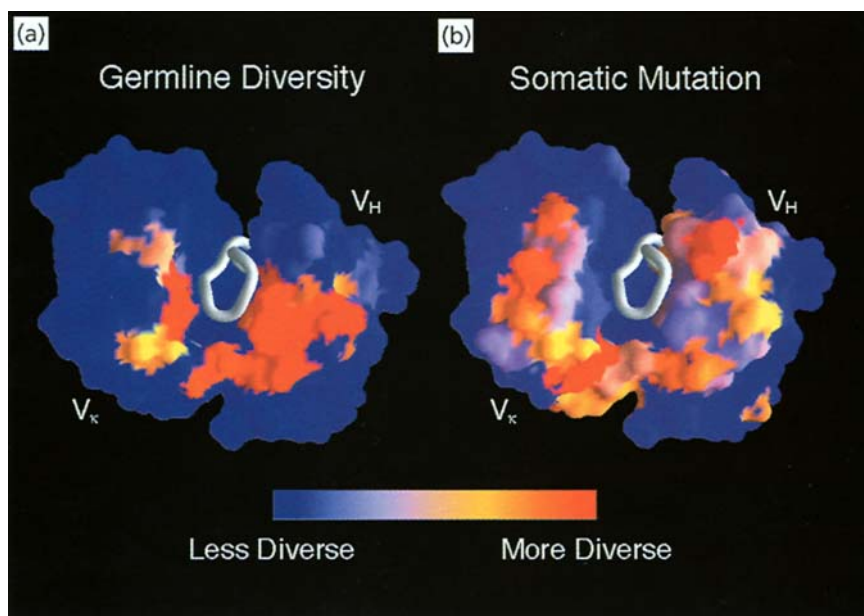


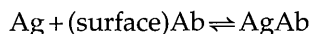
Figure 10.19. An 'antigen's eye view' of sequence diversity in human antibodies. The sequence diversity has been plotted on a scale of blue (more conserved) to red (more diverse). The V_H domain is on the right and the V_K domain on the left in both pictures. (a) Germ-line diversity prior to somatic hypermutation is focused at the center of the antigen-binding site. (b) Somatic hypermutation spreads diversity to regions at the periphery of the binding site that are highly conserved in the germ-line V gene repertoire.

Somatic hypermutation is therefore complementary to germ-line diversity. The V_H CDR3, which lies at the center of the antigen-binding site, was not included in this analysis and therefore is shown in gray as a loop structure. The end of the V_K CDR3 (also excluded) lies at the center of the binding site and is not visible in this representation. (Reproduced with kind permission from Tomlinson I.M. *et al.* (1996) *Journal of Molecular Biology* 256, 813.)

FACTORS AFFECTING ANTIBODY AFFINITY IN THE IMMUNE RESPONSE

The effect of antigen dose

Other things being equal, the binding strength of an antigen for the surface receptor of a B-cell will be determined by the affinity constant of the reaction:



and the reactants will behave according to the law of mass action (cf. p. 86).

It may be supposed that, when a sufficient number of antigen molecules are bound to the receptors on the cell surface and processed for presentation to T-cells, the lymphocyte will be stimulated to develop into an antibody-producing clone. When only small amounts of antigen are present, only those lymphocytes with high affinity receptors will be able to bind sufficient antigen for stimulation to occur and their daughter cells will, of course, also produce high affinity antibody. Consideration of the antigen–antibody equilibrium equation will show that, as the concentration of antigen is increased, even antibodies with relatively low affinity will bind more antigen; therefore, at high doses of antigen, the lymphocytes with lower affinity receptors will also be stimulated and, as may be seen from figure 10.20, these are more abundant than those with receptors of high affinity. Furthermore, there is a strong possibility that cells with the highest affinity will bind so much antigen as to become tolerized (cf. p. 240). Thus, in summary, low amounts of antigen produce high affinity antibodies, whereas high antigen concentrations give rise to an antiserum with low to moderate affinity.

Maturation of affinity

In addition to being brisker and fatter, secondary responses tend to be of higher affinity. There are probably two main reasons for this maturation of affinity after primary stimulation. First, once the primary response gets under way and the antigen concentration declines to low levels, only successively higher affinity cells will bind sufficient antigen to maintain proliferation. Second, at this stage the cells are mutating madly in the germinal centers, and any mutants with an adventitiously higher affinity will bind well to antigen on follicular dendritic cells and be positively selected for by its persistent clonal expansion. Modification of antibody specificity by somatic point mutations allows gradual diversification on which positive selection for affinity can act during clonal expansion.

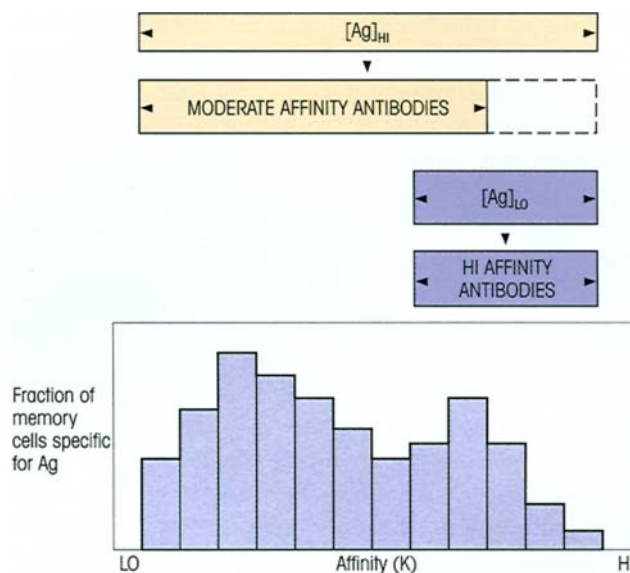


Figure 10.20. Relationship of antigen concentration to affinity of antibodies produced. Low concentrations of antigen ($[Ag]_{LO}$) bind to and permit stimulation of a range of high affinity memory cells and the resulting antibodies are of high affinity. High doses of antigen ($[Ag]_{HI}$) are able to bind sufficiently to the low affinity cells and thereby allow their stimulation, whilst the highest affinity cells may bind an excess of antigen and be tolerized (dashed line); the resulting antiserum will have a population of low to moderate affinity antibodies.

It is worth noting that responses to thymus-independent antigens, which have poorly developed memory with very rare mutations, do not show this phenomenon of affinity maturation. Overall, the ability of Th to facilitate responses to nonpolymeric, nonpolyclonally activating antigens, to induce expansive clonal proliferation, to effect class switching and, lastly, to fine-tune responses to higher affinity has provided us with bigger, better and more flexible immune responses.

MEMORY CELLS

Antibodies encoded by unmutated germ-line genes represent a form of evolutionary memory, in the sense that they tend to include specificities for commonly encountered pathogens which appear in the so-called 'natural antibody' fraction of serum. Memory acquired during the adaptive immune response requires contact with antigen and expansion of antigen-specific memory cells, as seen for example in the 20-fold increase in cytotoxic T-cell precursors after immunization of females with the male H-Y antigen.

Memory of early infections such as measles is long-lived and the question arises as to whether the memory cells are long-lived or are subject to repeated antigen

stimulation from persisting antigen or subclinical reinfection. Fanum in 1847 described a measles epidemic on the Faroe Islands in the previous year in which almost the entire population suffered from infection except for a few old people who had been infected 65 years earlier. While this evidence favors the long half-life hypothesis, memory function of B-cells transferred to an irradiated syngeneic recipient is lost within a month unless antigen is given or the donor is transgenic for the *bcl-2* gene (remember that signals in the germinal center which prevent apoptosis of centrocytic B-cells also upregulate *bcl-2* expression). It is envisaged that B-cell memory is a dynamic state in which survival of the memory cells is maintained by recurrent signals from follicular dendritic cells in the germinal centers, the only long-term repository of antigen.

Recent evidence strongly suggests that memory T-cells can, however, persist in the absence of antigen. T-cells isolated from mice several months after they were immunized with lymphocytic choriomeningitis virus (LCMV) were transferred into two groups of genetically modified mice which lacked endogenous T-cells, one of the groups additionally lacking MHC class I expression. T-cells were parked in these mice for 10 months and then analysed *in vitro*. Functional virus-specific CD8⁺ CTLs were still present in both groups of mice, and in similar numbers, even though those from the class I⁻ mice could not have had antigen presented to their TCR. Indeed, these memory T-cells undergo antigen- and MHC-independent proliferation *in vivo*, their numbers controlled, at least in part, by a balance between proliferation-inducing signals from IL-15 and cell death-inducing signals from IL-2 released in the local environment, both cytokines binding to the IL-2R β chain (cf. figure 10.2). Other recent findings indicate that helper T-cell memory also does not require the continued presence of antigen or MHC and, at least in some cases, Th memory is maintained in the absence of cell division.

However, we should not lose sight of the fact that, while these experiments in transgenic and knockout animals clearly demonstrate that immunological memory *can* be maintained in the absence of antigen, usually antigen persists as complexes on follicular dendritic cells. Therefore, there is the potential for antigen-presenting cells within the germinal center to capture and process this complexed antigen and then present it to memory T-cells. Some evidence, again recent (yes, this is a very rapidly moving area of immunology!), suggests that it is a type of dendritic cell, and not the germinal center B-cells, that may subserve this function.

The memory population is not simply an expansion of corresponding naive cells

In general, memory cells are more readily stimulated by a given dose of antigen because they have a higher affinity. In the case of B-cells, we have been satisfied by the evidence linking mutation and antigen selection to the creation of high affinity memory cells within the germinal center of secondary lymph node follicles. The receptors for antigen on memory T-cells also have higher affinity but, since they do not undergo significant somatic mutation during the priming response, it would seem that cells with **pre-existing receptors of relatively higher affinity in the population of naive cells proliferate selectively through preferential binding to the antigen.**

Intuitively one would not expect to improve on affinity to the same extent that somatic hypermutation can achieve for the B-cells, but nonetheless memory T-cells augment their binding avidity for the antigen-presenting cell through increased expression of accessory adhesion molecules, CD2, LFA-1, LFA-3 and ICAM-1. Since several of these molecules also function to enhance signal transduction, the memory T-cell is more readily triggered than its naive counterpart. Indeed, memory cells enter cell division and secrete cytokines more rapidly than naive cells, and there is some evidence that they may secrete a broader range of cytokines than do naive cells.

A phenotypic change in the isoform of the leukocyte common antigen CD45R, derived by differential splicing, allows some distinction to be made between naive and memory cells. Expression of CD45RA has been used as a marker of naive T-cells and of CD45RO as a marker of memory cells capable of responding to recall antigens. However, most of the features associated with the CD45RO subset are in fact manifestations of **activated cells** and CD45RO cells can revert to the CD45RA phenotype. Memory cells, perhaps in the absence of antigenic stimulation, may therefore lose their activated status and join a resting pool. Another marker used for differentiating naive from memory cells takes one step back on the CD ladder and utilizes differences in the relative expression of the adhesion molecule CD44; naive T-cells seem to express low levels of CD44 whilst memory T-cells express high levels.

It has been proposed by Lanzavecchia and colleagues that the CCR7 chemokine receptor allows a distinction to be made between CCR7⁺ 'central memory' T-cells, which differentiate from naive T-cells, and CCR7⁻ 'effector memory' T-cells, which subsequently arise from the central memory T-cells (figure 10.21). Both populations are long-lived. The central

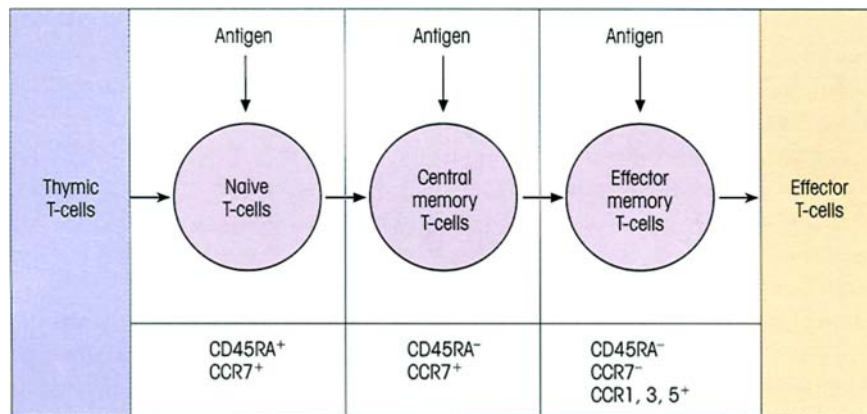


Figure 10.21. Central and effector memory T-cells. Naive T-cells bear the CD45RA splice variant of the CD45 molecule and are attracted from the thymus into secondary lymphoid tissue under the influence of CCR7-binding chemokines such as CCL19 (MIP-3 β) and CCL21 (6CKine/SLC). Upon encounter with antigen, some of these cells become effectors of the primary immune response, whilst others differentiate into central memory T-cells which retain the CCR7 chemokine receptor but lose expression of CD45RA. Subsequent re-encounter with antigen will push these cells into the effector

memory compartment with replacement of CCR7 by other chemokine receptors such as CCR1, CCR3 and CCR5. This changes the homing characteristics of these cells which can now relocate as cytokine-secreting or cytotoxic T-cells to inflammatory sites under the influence of a number of chemokines including CCL3 (MIP-1 α), CCL4 (MIP-1 β) and CCL5 (RANTES) (see table 10.3). Note that whilst the activation and subsequent differentiation of these cells is dependent on antigen, both central memory and effector memory T-cells are thought to be long-lived in the absence of antigen.

memory cells provide a clonally expanded pool of antigen-primed cells which can travel to secondary lymphoid organs under the influence of the CCL21 (SLC) chemokine (cf. table 10.3) and, following re-encounter with antigen, can stimulate dendritic cells, help B-cells and generate effector cells. In contrast, effector memory T-cells possess CCR1, CCR3 and CCR5 receptors for proinflammatory chemokines and constitute tissue-homing cells which mediate inflammatory reactions or cytotoxicity.

Virgin B-cells lose their surface IgM and IgD and switch receptor isotype on becoming memory cells, and the differential expression of these surface

markers has greatly facilitated the separation of B- and T-cells into naive and memory populations for further study. The costimulatory molecules B7.1 (CD80) and B7.2 (CD86) are rapidly upregulated on memory B-cells, and the potent ability of these cells to present antigen to T-cells could well account for the brisk and robust nature of secondary responses. A scheme similar to that outlined in figure 10.21 for T-cells may also exist for the B-lymphocyte compartment, with an initial population of memory cells possessing the B220 marker developing into B220⁻ memory B-cells which then go on to generate antibody-secreting effector cells.

SUMMARY

A succession of genes are upregulated by T-cell activation

- Within 15–30 minutes, genes for transcription factors concerned in the progression G0 to G1 and in the control of IL-2 are expressed.
- Up to 14 hours, cytokines and their receptors are expressed.
- Later, a variety of genes related to cell division and adhesion are upregulated.

Cytokines act as intercellular messengers

- Cytokines act transiently and usually at short range,

although circulating IL-1 and IL-6 can mediate release of acute phase proteins from the liver.

- They act through surface receptors belonging to six structural families.
- Cytokine-induced dimerization of individual subunits of the main (hematopoietin) receptor family activates protein tyrosine kinases, including JAKs, and leads to phosphorylation and activation of STAT transcription factors.
- Cytokines are pleiotropic, i.e. have multiple effects in the general areas of: (i) control of lymphocyte

(continued p. 198)